Uncommon Molecular Electronic Configurations

Theory, derivations and verification of Gryn'ova, Coote & Corminboeuf (2015)

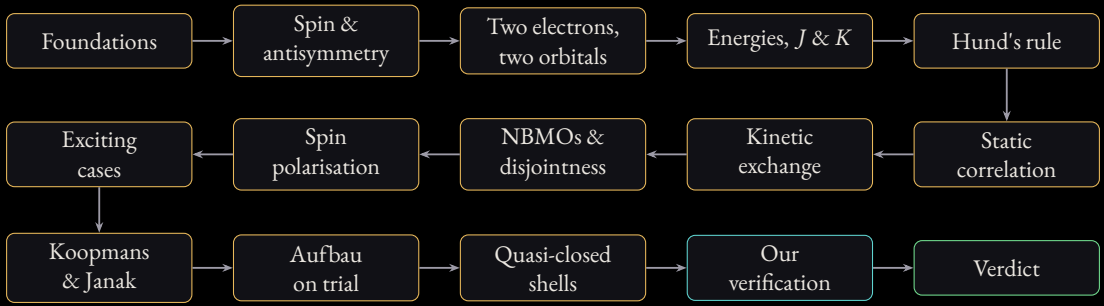
Shuvam Banerji Seal

CH5103 Term Paper

October 22, 2026

The journey

from the Schrödinger equation to a sign error in print



Every equation is derived step by step; every step was checked by SymPy, PySCF or Lean.

How to read these slides

The paper says
a statement quoted from the review

Verified
how we tested it (script / proof)

Assumptions
what a derivation takes for granted

Erratum
where the paper got it wrong

Result
what the derivation proves

VERIFIED ERROR OVERSTATED PARTIAL

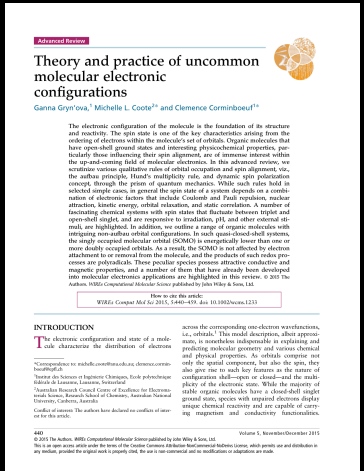
Atomic units throughout: $\hbar = m_e = e = 4\pi\varepsilon_0 = 1$, energies in hartree ($E_\mathrm{h}$).

Part I

The paper

I

The review



Advanced Review

Theory and practice of uncommon molecular electronic configurations

Ganna Gryn'ova,[1] Michelle L. Coote[2]* and Clemence Corminboeuf[1]*

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How to cite this article:
WIREs Comput Mol Sci 2015, 5:440–459. doi: 10.1002/wcms.1233

INTRODUCTION

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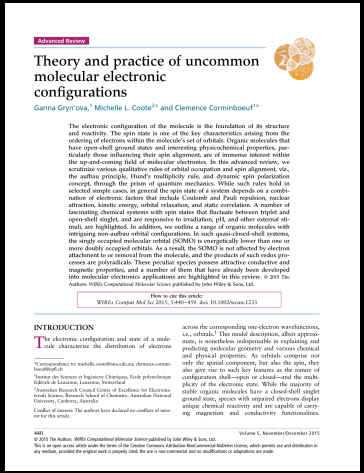
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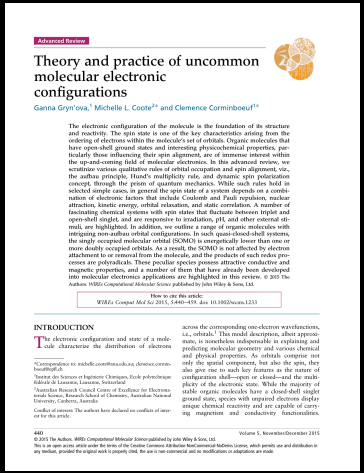
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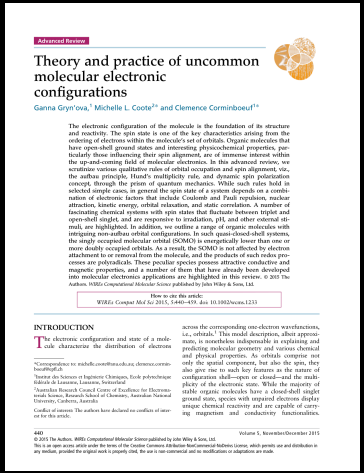
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The review



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- part theory (what decides a spin state?)
- part catalogue (molecules that break the textbook rules)

Two uncommon configurations

the two halves of the paper

---

1. FLUCTUATING MULTIPLICITY



triplet — open-shell singlet

two (nearly) degenerate orbitals, two electrons:
parallel or *antiparallel*?

2. NON-AUFBAU ORDERING



aufbau — quasi-closed shell

the singly occupied orbital lies *below*
a doubly occupied one

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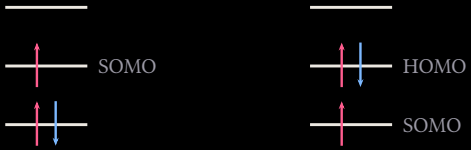


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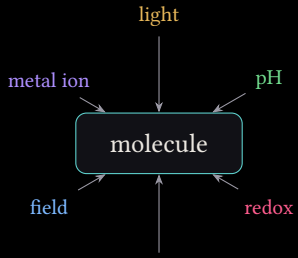
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Oxidation then makes a diradical.

Why anyone cares

The paper says

> "species with unpaired electrons display unique chemical reactivity and are capable of carrying magnetism and conductivity functionalities" (p. 440)



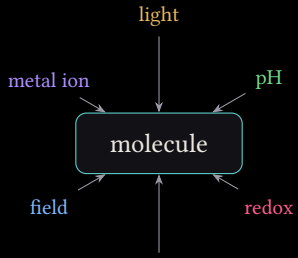
stimuli that switch spin states
(Figures 7–9 of the review)

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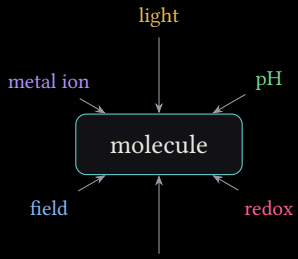
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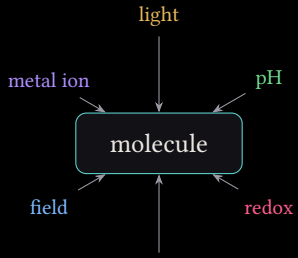
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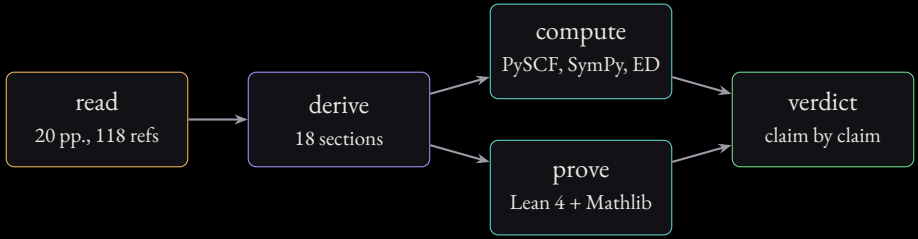
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- spin valves, switches, field-effect transistors, logic
- wanted: a set spin state that a stimulus can flip
- but few radicals are persistent, crystallisable *and* not Mott insulators

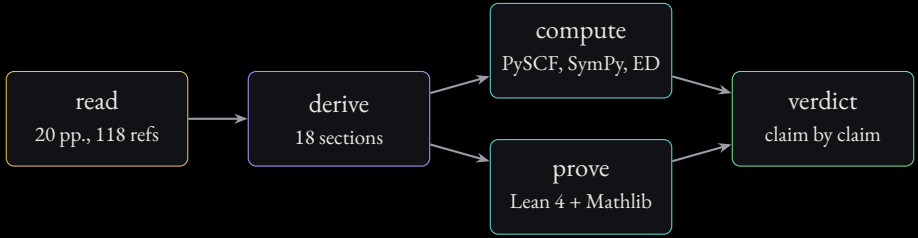


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What we did with it

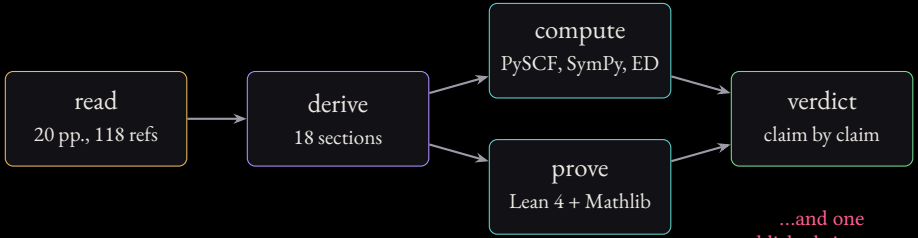


What we did with it



19 tests (T1–T17)
17 machine-checked theorems

What we did with it



Part II

Foundations

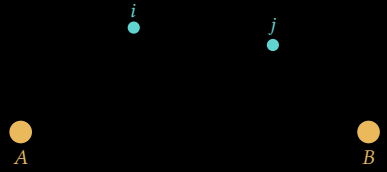
II

Where everything starts

the molecular Schrödinger equation

$$\hat{H}_{\text{mol}}\,\Psi = E\,\Psi$$

$$\hat{H}_{\text{mol}} = \underbrace{-\sum_A \frac{\nabla_A^2}{2M_A}}\ \underbrace{-\sum_i \frac{\nabla_i^2}{2}}\ \underbrace{-\sum_{i,A} \frac{Z_A}{r_{iA}}}\ \underbrace{+\sum_{i<j} \frac{1}{r_{ij}}}\ \underbrace{+\sum_{A<B} \frac{Z_A Z_B}{R_{AB}}}$$

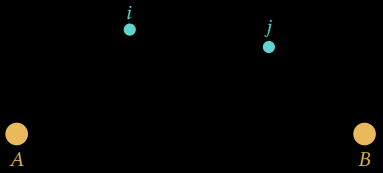


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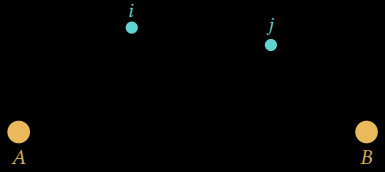


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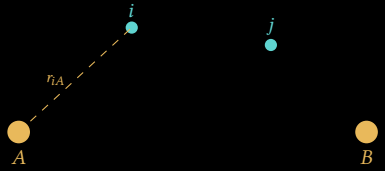


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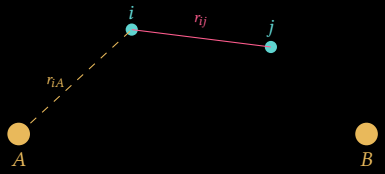


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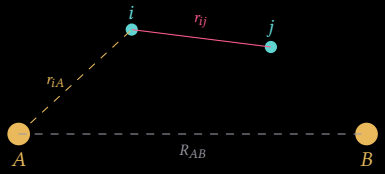


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Born--Oppenheimer: freeze the nuclei

Assumptions

$M_A \gg m_e = 1$: electrons adjust instantly to nuclear positions; treat $\{\mathbf{R}_A\}$ as parameters.

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1. drop the nuclear kinetic energy $-\sum_A \nabla_A^2 / 2M_A \to 0$
2. nuclear repulsion becomes a constant $V_{nn}(\mathbf{R})$

Born--Oppenheimer: freeze the nuclei

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1. drop the nuclear kinetic energy — $-\sum_A \nabla_A^2 / 2M_A \to 0$
2. nuclear repulsion becomes a constant — $V_{nn}(\mathbf{R})$
3. what is left is the electronic Hamiltonian

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Note: $\hat{H}$ contains no spin. Spin will enter only through the Pauli principle.

The variational principle

why "lower energy" means "better wavefunction"

Assume: exact eigenstates $\hat{H}\Psi_n = E_n\Psi_n$, orthonormal, complete, $E_0 \le E_1 \le \cdots$

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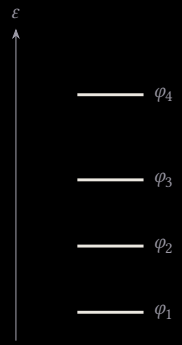
Result

$E[\tilde{\Psi}] \ge E_0$ for every normalised $\tilde{\Psi}$; Hartree–Fock and CI rest on it.

Orbitals: the independent-electron picture

Independent-particle model

replace $\sum_{i<j} 1/r_{ij}$ by an *average* field felt by each electron

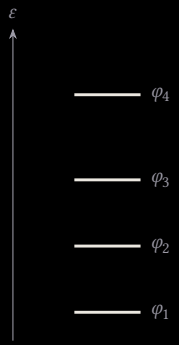


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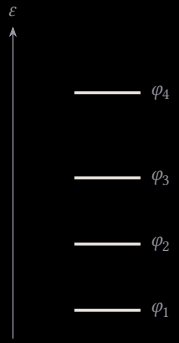
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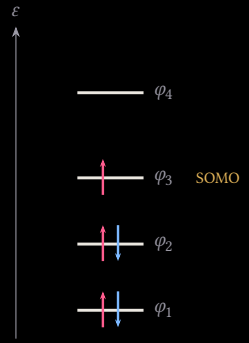
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orbital = one-electron wavefunction; its *energy* ε_p and its *occupation* build the configuration.



Box 1: ``Much ado about orbitals''

The paper says

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Keep this in mind: it returns when ROHF orbital energies turn out to be arbitrary.

Part III

Spin & antisymmetry

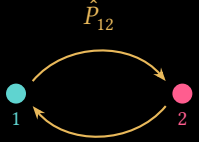
III

Swapping two electrons

why the exchange eigenvalue is ± 1

Define $\hat{P}_{12}\Psi(x_1, x_2) = \Psi(x_2, x_1), \quad x = (\mathbf{r}, \sigma).$

(1) $\hat{P}_{12}^2 = \hat{1}$ swap twice



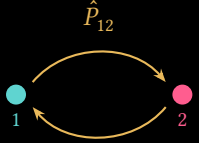
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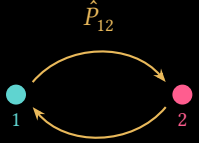
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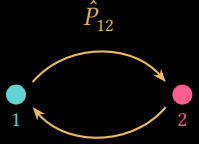
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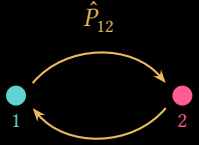
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$\lambda = -1$

electrons (fermions)

The paper says

> "the wavefunction for a system of fermions must be antisymmetric with respect to the interchange of any two particles" (p. 441)

Slater determinants

antisymmetry built in

$$D[\chi_1, \chi_2] = \begin{vmatrix} \chi_1(1) & \chi_2(1) \\ \chi_1(2) & \chi_2(2) \end{vmatrix} = \chi_1(1)\chi_2(2) - \chi_2(1)\chi_1(2)$$

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Result

"No two electrons with the same four quantum numbers" is a *consequence* of antisymmetry inside the orbital model, not a separate law.

Spin operators for two electrons

One electron ($s = \frac{1}{2}$):

$$\hat{s}_z\alpha = +\tfrac{1}{2}\alpha, \qquad \hat{s}_z\beta = -\tfrac{1}{2}\beta$$
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$$\boxed{\hat{S}^2 = \hat{S}_-\hat{S}_+ + \hat{S}_z^2 + \hat{S}_z}$$

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$$\hat{S}_+\alpha_1\alpha_2 = 0$$
$$\hat{S}_z\alpha_1\alpha_2 = (\tfrac{1}{2} + \tfrac{1}{2})\alpha_1\alpha_2 = \alpha_1\alpha_2$$
$$\hat{S}^2\alpha_1\alpha_2 = (0 + 1 + 1)\,\alpha_1\alpha_2$$
$$= 2\,\alpha_1\alpha_2$$

$S(S+1) = 2 \Rightarrow S = 1,\ M_S = +1$

$\beta_1\beta_2$

$$\hat{S}_+\beta_1\beta_2 = \alpha_1\beta_2 + \beta_1\alpha_2$$
$$\hat{S}_-(\alpha_1\beta_2 + \beta_1\alpha_2) = \beta_1\beta_2 + \beta_1\beta_2$$
$$(\hat{S}_z^2 + \hat{S}_z)\beta_1\beta_2 = (1-1)\beta_1\beta_2 = 0$$
$$\hat{S}^2\beta_1\beta_2 = 2\,\beta_1\beta_2$$

$S = 1,\ M_S = -1$

$\hat{S}^2$ on the four spin products (2/2)

$\alpha_1\beta_2$ and $\beta_1\alpha_2$ ($M_S = 0$; neither is an eigenfunction on its own)

$$\hat{S}_+\alpha_1\beta_2 = \alpha_1\alpha_2$$

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$$\hat{S}_+\alpha_1\beta_2 = \alpha_1\alpha_2 \qquad \hat{S}_-\alpha_1\alpha_2 = \beta_1\alpha_2 + \alpha_1\beta_2$$

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Add and subtract:

$$\hat{S}^2(\alpha_1\beta_2 \pm \beta_1\alpha_2) = (1 \pm 1)\,(\alpha_1\beta_2 \pm \beta_1\alpha_2)$$

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Add and subtract:

$$\hat{S}^2(\alpha_1\beta_2 \pm \beta_1\alpha_2) = (1 \pm 1)(\alpha_1\beta_2 \pm \beta_1\alpha_2)$$

$+$: eigenvalue 2 $\Rightarrow S = 1$ $\quad$ $-$: eigenvalue 0 $\Rightarrow S = 0$

Triplet and singlet spin functions

The four two-electron spin states

$$\Theta_{1,+1} = \alpha_1\alpha_2$$
$$\Theta_{1,0} = \frac{1}{\sqrt{2}}(\alpha_1\beta_2 + \beta_1\alpha_2)$$
$$\Theta_{1,-1} = \beta_1\beta_2$$
$$\Theta_{0,0} = \frac{1}{\sqrt{2}}(\alpha_1\beta_2 - \beta_1\alpha_2)$$

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- total must be antisymmetric, so

	space	spin
triplet	anti	sym
singlet	sym	anti

Triplet and singlet spin functions

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✓ T1: $\hat{S}^2, \hat{S}_z$ built as 4×4 matrices in SymPy; all eigen-relations exact.

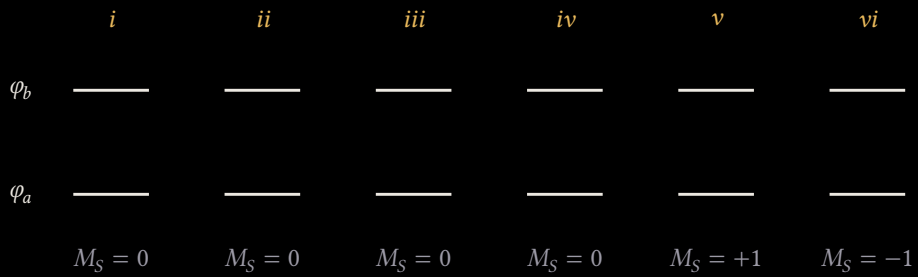
Part IV

Two electrons, two orbitals

IV

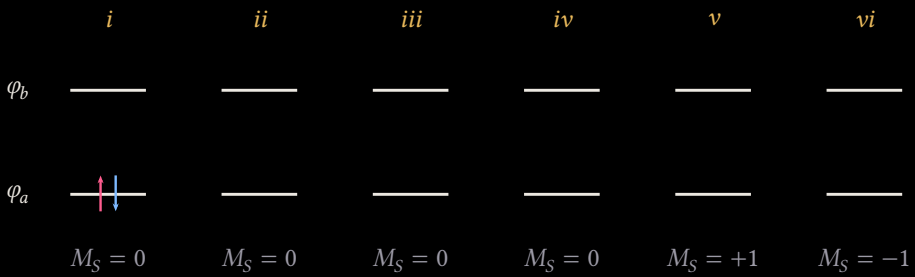
Six ways to place two electrons

Figure 1 of the review, rebuilt



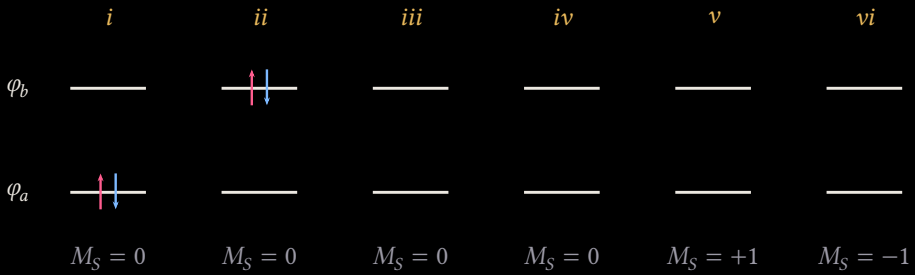
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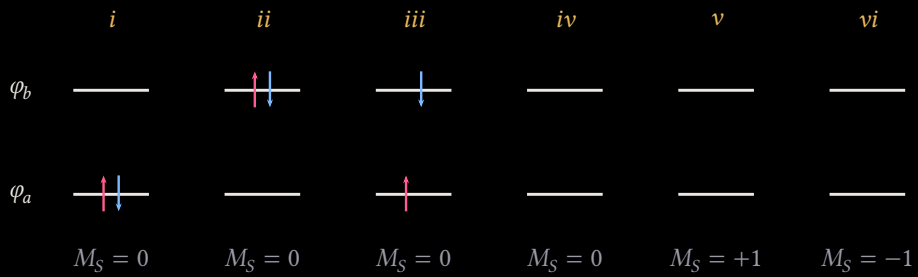
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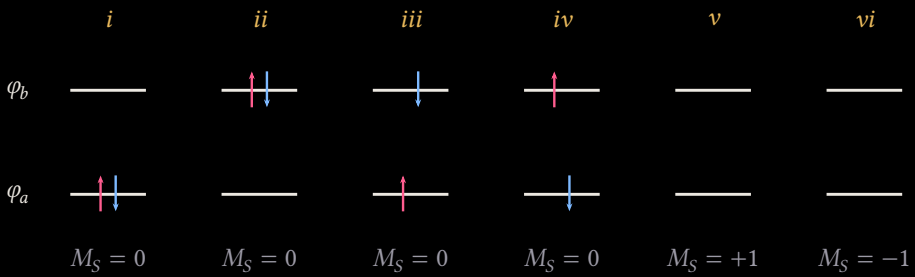
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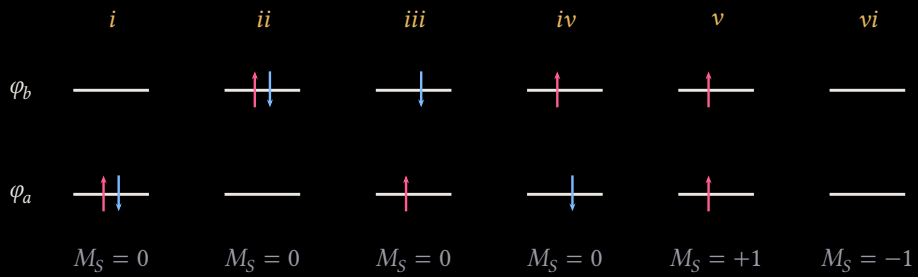
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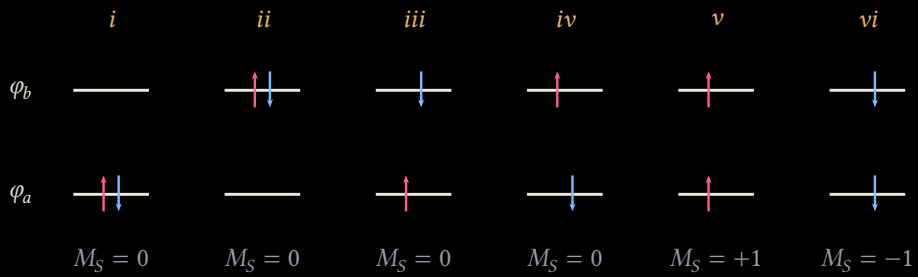
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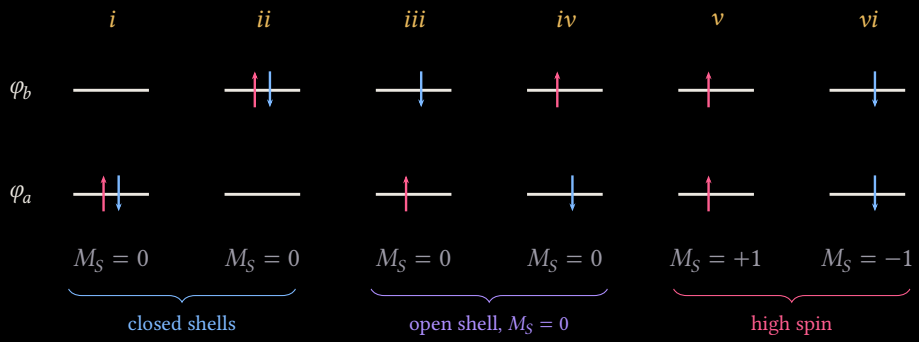
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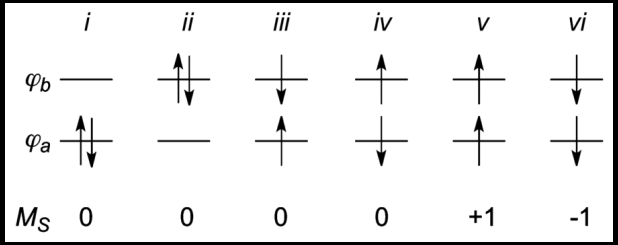
Six ways to place two electrons

Figure 1 of the review, rebuilt



four spin orbitals $\varphi_a\alpha, \varphi_a\beta, \varphi_b\alpha, \varphi_b\beta \Rightarrow \binom{4}{2} = 6$ determinants

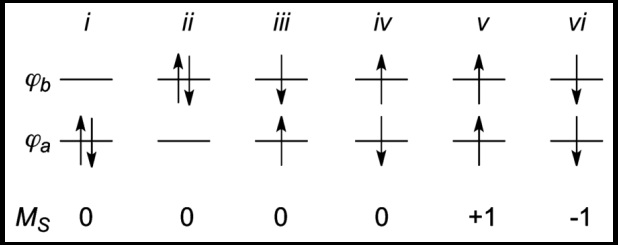
The paper's Figure 1



The paper says

"Various arrangements (determinants) of two electrons in two molecular orbitals and the corresponding values of the secondary spin quantum number."

The paper's Figure 1



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"Various arrangements (determinants) of two electrons in two molecular orbitals and the corresponding values of the secondary spin quantum number."

Matches our reconstruction. Now write each one out.

Writing the determinants out

shorthand: $A \equiv \varphi_a(1)\varphi_b(2)$, $B \equiv \varphi_b(1)\varphi_a(2)$

	spin orbitals	$D = \chi_1(1)\chi_2(2) - \chi_2(1)\chi_1(2)$
i	$\varphi_a\alpha,\ \varphi_a\beta$	$\varphi_a(1)\varphi_a(2)\,(\alpha_1\beta_2 - \beta_1\alpha_2)$
ii	$\varphi_b\alpha,\ \varphi_b\beta$	$\varphi_b(1)\varphi_b(2)\,(\alpha_1\beta_2 - \beta_1\alpha_2)$

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iii	$\varphi_a\alpha,\ \varphi_b\beta$	$A\,\alpha_1\beta_2 - B\,\beta_1\alpha_2$

e.g. iii: $\chi_1 = \varphi_a\alpha,\ \chi_2 = \varphi_b\beta$: $\varphi_a(1)\alpha_1\,\varphi_b(2)\beta_2 - \varphi_b(1)\beta_1\,\varphi_a(2)\alpha_2$

Writing the determinants out

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iv	$\varphi_a\beta,\ \varphi_b\alpha$	$A\,\beta_1\alpha_2 - B\,\alpha_1\beta_2$
v	$\varphi_a\alpha,\ \varphi_b\alpha$	$(A - B)\,\alpha_1\alpha_2$
vi	$\varphi_a\beta,\ \varphi_b\beta$	$(A - B)\,\beta_1\beta_2$

e.g. iii: $\chi_1 = \varphi_a\alpha,\ \chi_2 = \varphi_b\beta$: $\varphi_a(1)\alpha_1\,\varphi_b(2)\beta_2 - \varphi_b(1)\beta_1\,\varphi_a(2)\alpha_2$

$M_S = \pm 1$: *one determinant each*

$$\frac{1}{\sqrt{2}} D_v = \underbrace{\frac{1}{\sqrt{2}}\left[\varphi_a(1)\varphi_b(2) - \varphi_b(1)\varphi_a(2)\right]}_{\textit{antisymmetric space}} \underbrace{\alpha_1\alpha_2}_{\Theta_{1,+1}}$$

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$$\frac{1}{\sqrt{2}} D_{vi} = \frac{1}{\sqrt{2}}\left[\varphi_a(1)\varphi_b(2) - \varphi_b(1)\varphi_a(2)\right] \beta_1\beta_2$$

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The paper says

"${}^3\Psi(M_s = \pm 1) = \frac{1}{\sqrt{2}}[\varphi_a(1)\varphi_b(2) - \varphi_b(1)\varphi_a(2)]\,[\alpha_1\alpha_2]$ ... i.e. they can be represented as single determinants v and vi" (p. 441)

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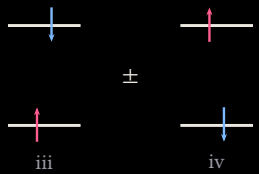
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VERIFIED exactly right: the high-spin triplet is single-reference.

$M_S = 0$: *iii and iv are not spin eigenfunctions*

$$D_{\text{iii}} = A \underbrace{\alpha_1\beta_2}_{\text{not } \Theta} - B \underbrace{\beta_1\alpha_2}_{\text{not } \Theta}$$

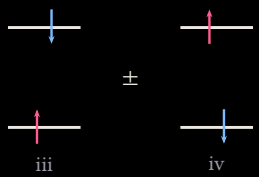


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$$D_{\text{iii}} = A \underbrace{\alpha_1\beta_2}_{\text{not } \Theta} - B \underbrace{\beta_1\alpha_2}_{\text{not } \Theta}$$

From the spin slides:

$$\hat{S}^2\alpha_1\beta_2 = \alpha_1\beta_2 + \beta_1\alpha_2 \neq \lambda\,\alpha_1\beta_2$$



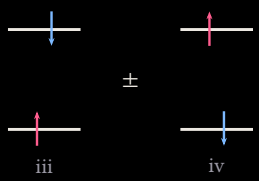
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From the spin slides:

$$\hat{S}^2\alpha_1\beta_2 = \alpha_1\beta_2 + \beta_1\alpha_2 \neq \lambda\,\alpha_1\beta_2$$

so D_{iii} is half singlet, half triplet.
Only combinations of iii and iv have definite spin.



The sum $D_{iii} + D_{iv}$

every term tracked

$$D_{\mathrm{iii}} + D_{\mathrm{iv}} = A\,\alpha_1\beta_2 - B\,\beta_1\alpha_2 + A\,\beta_1\alpha_2 - B\,\alpha_1\beta_2 \qquad \text{substitute}$$

The sum $D_{iii} + D_{iv}$

every term tracked

$$
\begin{aligned}
D_{\mathrm{iii}} + D_{\mathrm{iv}} &= A\,\alpha_1\beta_2 - B\,\beta_1\alpha_2 + A\,\beta_1\alpha_2 - B\,\alpha_1\beta_2 && \text{substitute} \\
&= A\,(\alpha_1\beta_2 + \beta_1\alpha_2) - B\,(\beta_1\alpha_2 + \alpha_1\beta_2) && \text{collect } A,\ B
\end{aligned}
$$

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every term tracked

$$
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&= (A - B)\,(\alpha_1\beta_2 + \beta_1\alpha_2) && \text{same bracket}
\end{aligned}
$$

The sum $D_{iii} + D_{iv}$

every term tracked

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&= (A - B)\,(\alpha_1\beta_2 + \beta_1\alpha_2) && \text{same bracket} \\
&= \underbrace{\left[\varphi_a(1)\varphi_b(2) - \varphi_b(1)\varphi_a(2)\right]}_{\text{antisymmetric space}}\ \underbrace{(\alpha_1\beta_2 + \beta_1\alpha_2)}_{\sqrt{2}\,\Theta_{1,0}}
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\end{aligned}
$$

Result

The *sum* of iii and iv is the $M_S = 0$ component of the triplet.

The difference $D_{iv} - D_{iii}$

$$D_{iv} - D_{iii} = A\,\beta_1\alpha_2 - B\,\alpha_1\beta_2 - A\,\alpha_1\beta_2 + B\,\beta_1\alpha_2 \qquad \textit{substitute}$$

The difference $D_{iv} - D_{iii}$

$$
\begin{aligned}
D_{iv} - D_{iii} &= A\,\beta_1\alpha_2 - B\,\alpha_1\beta_2 - A\,\alpha_1\beta_2 + B\,\beta_1\alpha_2 && \textit{substitute} \\
&= A\,(\beta_1\alpha_2 - \alpha_1\beta_2) + B\,(\beta_1\alpha_2 - \alpha_1\beta_2) && \textit{collect}
\end{aligned}
$$

The difference $D_{iv} - D_{iii}$

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\begin{aligned}
D_{iv} - D_{iii} &= A\,\beta_1\alpha_2 - B\,\alpha_1\beta_2 - A\,\alpha_1\beta_2 + B\,\beta_1\alpha_2 && \textit{substitute} \\
&= A\,(\beta_1\alpha_2 - \alpha_1\beta_2) + B\,(\beta_1\alpha_2 - \alpha_1\beta_2) && \textit{collect} \\
&= (A + B)\,(\beta_1\alpha_2 - \alpha_1\beta_2)
\end{aligned}
$$

The difference $D_{iv} - D_{iii}$

$$
\begin{aligned}
D_{iv} - D_{iii} &= A\,\beta_1\alpha_2 - B\,\alpha_1\beta_2 - A\,\alpha_1\beta_2 + B\,\beta_1\alpha_2 && \textit{substitute} \\
&= A\,(\beta_1\alpha_2 - \alpha_1\beta_2) + B\,(\beta_1\alpha_2 - \alpha_1\beta_2) && \textit{collect} \\
&= (A + B)\,(\beta_1\alpha_2 - \alpha_1\beta_2) \\
&= \underbrace{\left[\varphi_a(1)\varphi_b(2) + \varphi_b(1)\varphi_a(2)\right]}_{\textit{symmetric space}} \underbrace{(\beta_1\alpha_2 - \alpha_1\beta_2)}_{-\sqrt{2}\,\Theta_{0,0}}
\end{aligned}
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The difference $D_{iv} - D_{iii}$

$$
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Result

The *difference* is the open-shell singlet.

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Result

$$^3\Psi_0 = \tfrac{1}{2}(D_{\text{iii}} + D_{\text{iv}})$$

$$^1\Psi_{\text{OSS}} = \tfrac{1}{2}(D_{\text{iv}} - D_{\text{iii}})$$

Now read the paper

spot the difference

requires multiple configurations. For example, the open-shell singlet wavefunction

$$^1\Psi_{(M_s=0)} = \frac{1}{\sqrt{2}}[\varphi_a(1)\varphi_b(2) + \varphi_b(1)\varphi_a(2)] * \frac{1}{\sqrt{2}}[\beta_1\alpha_2 - \alpha_1\beta_2]$$

is a combination of two determinants, *iii* and *iv*:

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Line 1 (product form) … then “is a combination of two determinants, iii and iv:”

Spotted it

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- green: product form, symmetric space × antisymmetric spin VERIFIED

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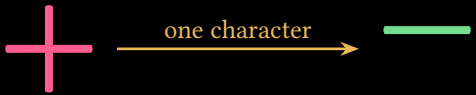
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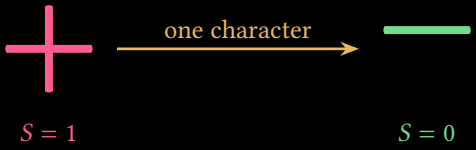
Erratum

should read $\frac{1}{2}(D_{\text{iv}} - D_{\text{iii}})$

Fluctuating multiplicity, live in print

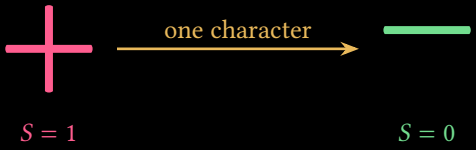


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The section is titled *"Fluctuating ground-state multiplicity"*...

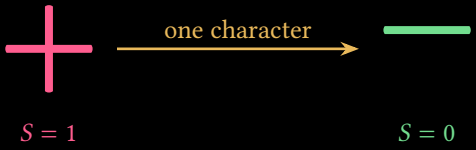
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...and between two lines the open-shell singlet fluctuates into a triplet.

Fluctuating multiplicity, live in print



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…and between two lines the open-shell singlet fluctuates into a triplet.

(It survived peer review and typesetting. Two referees with parallel spins, presumably: the Fermi hole kept them from ever meeting the error.)

Checked twice: SymPy and Lean

SymPy (T1)

```
3Psi(0)  = (iii+iv)/2          : True
1Psi_OSS = (iv-iii)/2          : True
paper's (iii+iv)/2 = OSS ?     : False
T1 PASS
```

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full symbolic expansion of both determinants

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full symbolic expansion of both determinants

Lean 4 + Mathlib (T17)

```
theorem sum_is_triplet (A B ab ba : R) :
    Diii A B ab ba + Div A B ab ba
      = (A - B) * (ab + ba) := by
  unfold Diii Div; ring

theorem erratum_plus_is_not_singlet :
    ¬ (∀ A B ab ba : ℤ,
      Diii A B ab ba + Div A B ab ba
        = (A + B) * (ba - ab)) := by
  intro h
  have := h 1 0 1 0
  simp [Diii, Div] at this
```

What the two-electron problem teaches

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The paper says

"their triplet state is therefore accurately described by a single reference determinant, however open-shell singlets require multiconfigurational (multireference) treatment."

What the two-electron problem teaches

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The paper says

"their triplet state is therefore accurately described by a single reference determinant, however open-shell singlets require multiconfigurational (multireference) treatment."

VERIFIED conclusion right, despite the typo

Part V

Energies: J and K



The integrals we need

$$h_{pq} = \int \varphi_p(\mathbf{r})\, \hat{h}\, \varphi_q(\mathbf{r})\, \mathrm{d}\mathbf{r}$$

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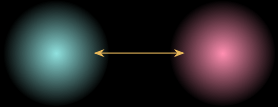
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$$J_{pq} = (pp|qq)$$

J: cloud $|\varphi_a|^2$ repels cloud $|\varphi_b|^2$



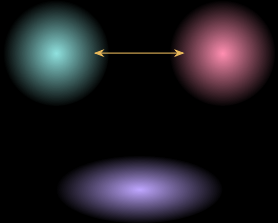
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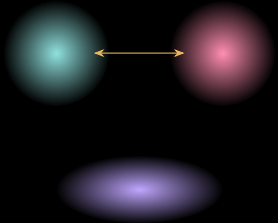
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chemists' notation; real orbitals ⇒
$(pq|rs) = (qp|rs) = (rs|pq) = \cdots$ (8-fold)

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Two elementary integrals

orthonormality does most of the work

Assumptions

real orthonormal orbitals $\langle \varphi_p | \varphi_q \rangle = \delta_{pq}$; $\hat{h}(1)$ acts only on electron 1

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electron 1 carries p, r; electron 2 carries q, s

Triplet energy, step by step (1/2)

$\Phi_T = \frac{1}{\sqrt{2}}(A - B)$, $\hat{H} = \hat{h}(1) + \hat{h}(2) + r_{12}^{-1}$, spin part normalised $\Rightarrow$ integrates to 1

One-electron part

$$\langle A|\hat{h}(1) + \hat{h}(2)|A\rangle = h_{aa} \cdot 1 + 1 \cdot h_{bb} = h_a + h_b$$

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$$\tfrac{1}{2}\big[(h_a + h_b) + (h_a + h_b) - 0 - 0\big] = \boxed{h_a + h_b}$$

Triplet energy, step by step (2/2)

Two-electron part

$$\langle A|r_{12}^{-1}|A\rangle = (aa|bb) = J_{ab} \qquad p, r = a;\ q, s = b$$

Triplet energy, step by step (2/2)

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$$\tfrac{1}{2}\big[J + J - K - K\big] = J_{ab} - K_{ab}$$

Result

$$E_T = h_a + h_b + J_{ab} - K_{ab}$$

Open-shell singlet and the gap

$\Phi_{\mathrm{OSS}} = \frac{1}{\sqrt{2}}(A + B)$: identical steps, but the cross terms enter with +

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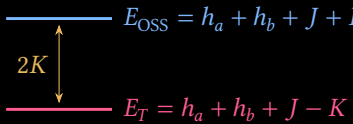
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The full two-electron, two-orbital Hamiltonian

basis $\{\Phi_{\mathrm{i}}, \Phi_{\mathrm{ii}}, \Phi_{\mathrm{OSS}}\}$ (singlets) + the triplet on its own

$$\mathbf{H}_S = \begin{pmatrix} 2h_a + J_{aa} & K_{ab} & \sqrt{2}\,(h_{ab} + (aa|ab)) \\ K_{ab} & 2h_b + J_{bb} & \sqrt{2}\,(h_{ab} + (bb|ba)) \\ \cdot & \cdot & h_a + h_b + J_{ab} + K_{ab} \end{pmatrix}, \qquad E_T = h_a + h_b + J_{ab} - K_{ab}$$

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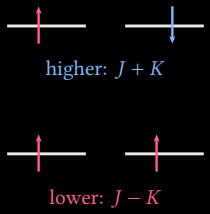
T2: BRUTE FORCE

5 random integral sets (positive-definite kernel) → PySCF full CI: singlet and triplet roots reproduce these closed forms to $2 \times 10^{-14}\,E_{\mathrm{h}}$.

Hund's rule at frozen orbitals

The paper says

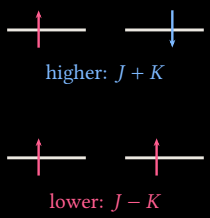
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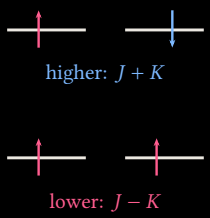
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Two things to prove: $K \geq 0$ (next section), and whether "same orbitals" is legitimate (it is not).

Part VI

Fermi holes

VI

Where can the two electrons be?

the pair density

$$P(\mathbf{r}_1, \mathbf{r}_2) = N(N-1) \sum_{\sigma_1 \sigma_2} \int |\Psi|^2 \, \mathrm{d}x_3 \cdots \mathrm{d}x_N$$

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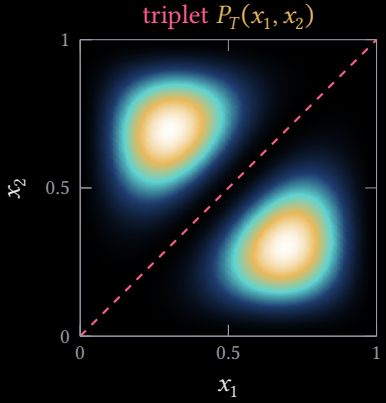
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The *Coulomb* hole (from $1/r_{12}$ itself) is absent here: it needs configuration mixing, i.e. correlation.

Seeing the Fermi hole

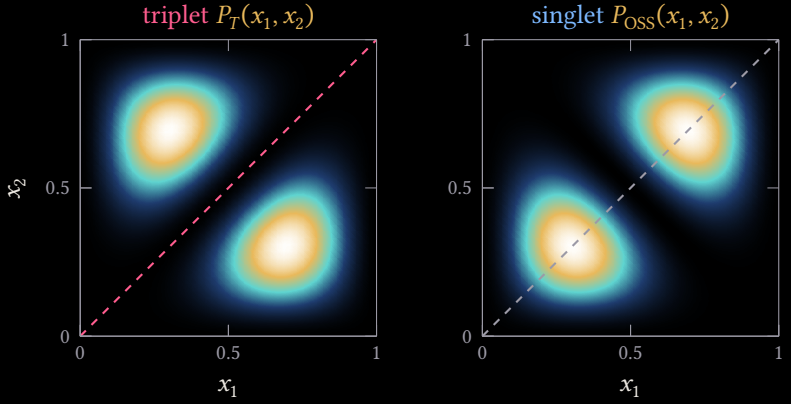
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Seeing the Fermi hole

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$K_{ab} \geq 0$: *a three-line proof*

overlap density $\rho(\mathbf{r}) = \varphi_a(\mathbf{r})\varphi_b(\mathbf{r})$ (real) $\quad\Rightarrow\quad K_{ab} = \iint \dfrac{\rho(1)\rho(2)}{r_{12}}$

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equality iff $\varphi_a\varphi_b \equiv 0$ (no common region of space) — this returns as *disjointness*

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T3

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so $E_T = h_a + h_b + J - K$ really is the lower of the two.

Part VII

Hund's rule

VII

Two explanations of the same rule

The paper says

> "Slater's original explanation … was therefore based upon less electron–electron Coulomb repulsion in the triplet state … However, it has been later shown that the electron–electron repulsion is actually greater in the high-spin state, and an alternative explanation — the so-called less screening rule — has been offered." (p. 442)

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Less screening (relaxed orbitals)

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Which one is right? The virial theorem decides.

The virial theorem by scaling (1/2)

Assumptions

an atom (nucleus at the origin); exact eigenstate Ψ, or any Ψ optimised with respect to scaling

$$\Psi_\eta(\mathbf{r}_1, \dots, \mathbf{r}_N) = \eta^{3N/2}\, \Psi(\eta\mathbf{r}_1, \dots, \eta\mathbf{r}_N) \qquad \text{squeeze } (\eta > 1) \text{ or stretch} \tag{1}$$

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$= \int |\Psi(\mathbf{s})|^2 \mathrm{d}^{3N} s = 1$

The virial theorem by scaling (1/2)

Assumptions

an atom (nucleus at the origin); exact eigenstate Ψ, or any Ψ optimised with respect to scaling

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the scaled function is still normalised, so it is a legal trial function

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Lean: `scaling_derivative`. Molecules: zero at equilibrium geometries.

Frozen orbitals cannot be the mechanism

Apply the virial theorem to *both* exact states and subtract:

$$\Delta E \equiv E_T - E_S = -\Delta\langle T\rangle = \tfrac{1}{2}\big(\Delta\langle V_{ne}\rangle + \Delta\langle V_{ee}\rangle\big)$$

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Hongo et al. (2004): “This assumption leaves the kinetic energy T unchanged for different spin states, which cannot be correct as each eigenstate must satisfy the virial theorem.”

Less screening, as inequalities

Assumptions

triplet lower ($\Delta E < 0$), both states virial, $\Delta V_{ee} > 0$ (what the calculations show)

(1) $\Delta T = -\Delta E > 0$ electrons more compressed

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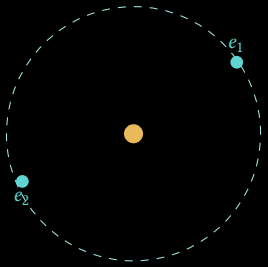
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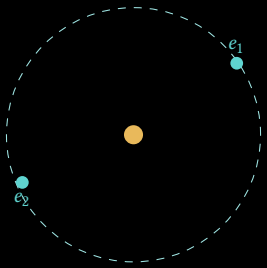
the nucleus wins: stronger attraction over-compensates stronger repulsion (Lean: `less_screening`)

The physical picture

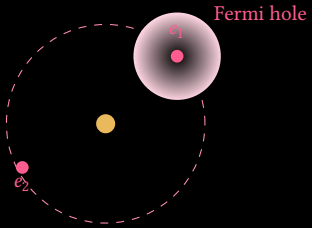


singlet: electrons may overlap,
each screens the nucleus

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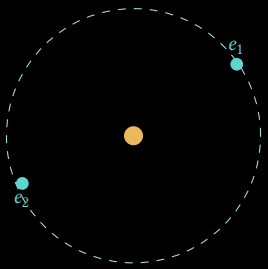


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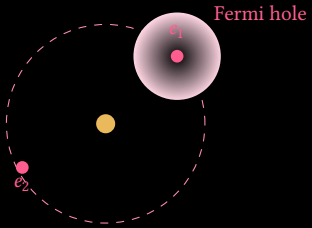


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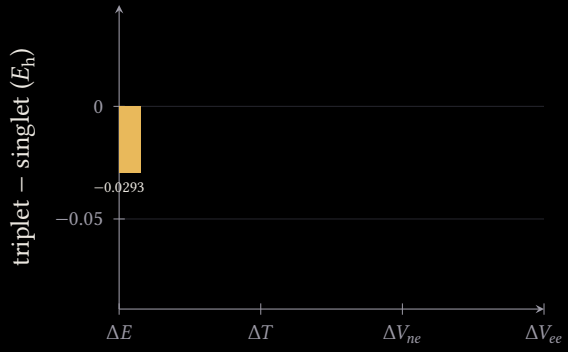


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contraction lowers V_{ne} • raises T • also raises V_{ee} (closer on average)

Test 1: helium 1s2s, solved exactly

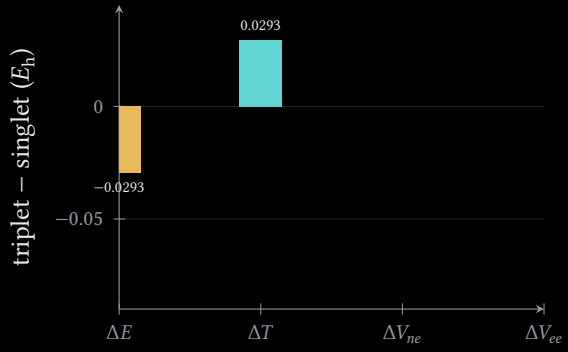
T4 --- $2\,^3S$ vs $2\,^1S$, 16s7p4d basis, scaled to satisfy the virial theorem



	$2\,^3S$	$2\,^1S$
E (ours)	−2.175212	−2.145865
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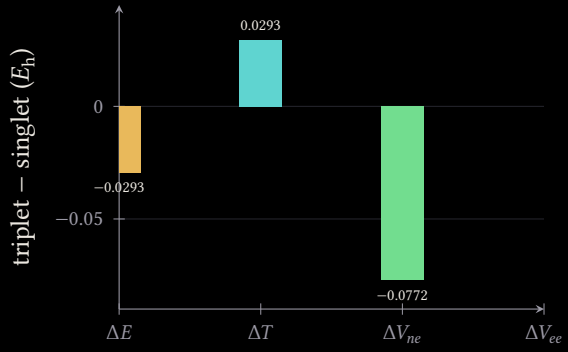
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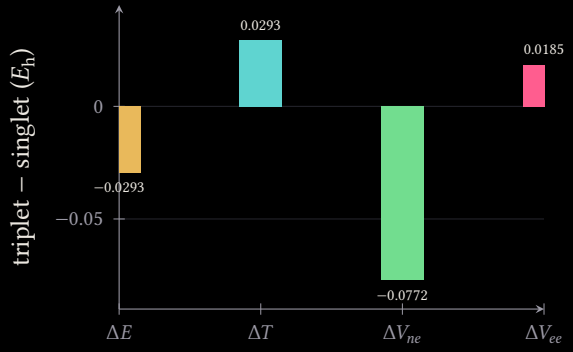
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$\Delta V_{ee} > 0$: Slater's mechanism fails; ΔV_{ne} pays for everything.

check:
$\frac{1}{2}(-0.0772 + 0.0185) = -0.0293 = \Delta E$

Test 2: the carbon atom (literature)

Hongo et al., J. Chem. Phys. 121, 7144 (2004), Table I --- read from the arXiv full text

$^3P - {}^1D$ (E_h)	ΔV_{ee}	ΔV_{en}	ΔT
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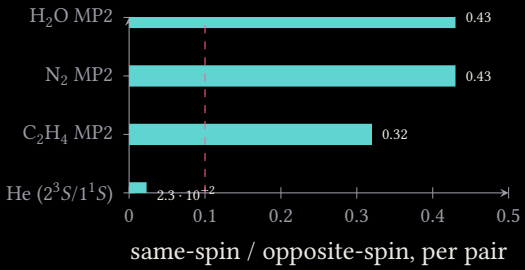
Result

The review's account (less screening, V_{ee} *greater* in the high-spin state) is right. VERIFIED

A side claim: same-spin correlation

The paper says

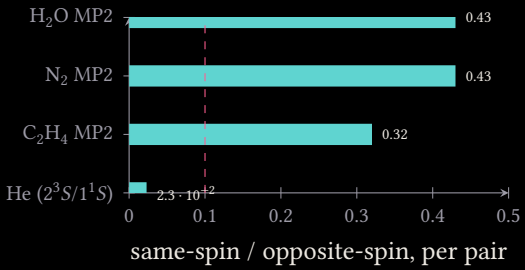
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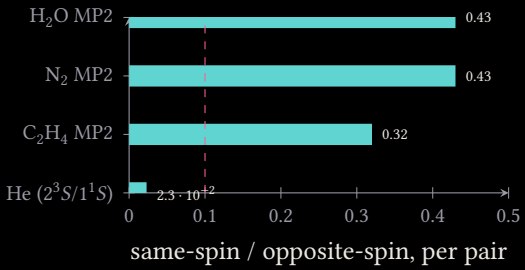


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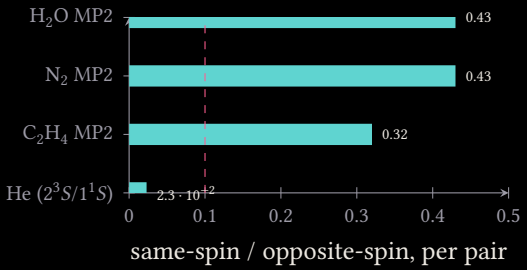
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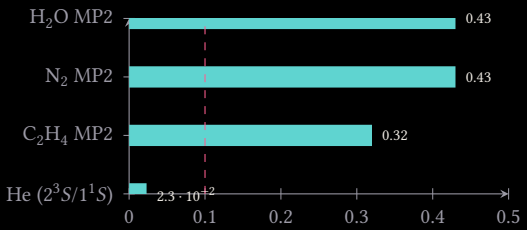
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OVERSTATED true for compact two-electron systems, overstated in general (T15)

Part VIII

Static correlation

VIII

Static versus dynamic correlation

The paper says

"The weaker the interaction between the two nonbonding molecular orbitals (MOs), the stronger the static correlation (Figure 2)." (p. 442)

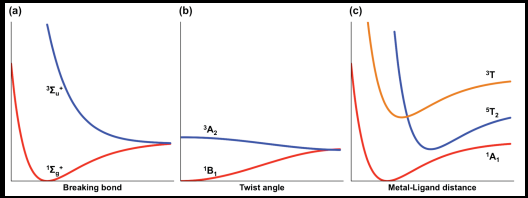


Fig. 2: (a) H_2 bond breaking (b) ethylene twist (c) metal–ligand distance

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dynamic: electrons dodging each other within one configuration

static: several configurations of comparable weight

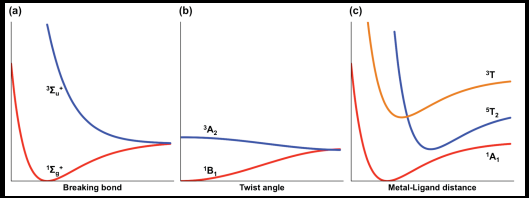


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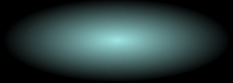
$\mathcal{H}_2$ in a minimal basis

one 1s function per atom, χ_A, χ_B, overlap S:

$$\sigma_g \equiv \varphi_1 = \frac{\chi_A + \chi_B}{\sqrt{2(1+S)}}$$

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σ_g (bonding)



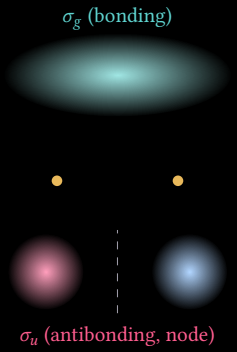
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φ_1 even, φ_2 odd under inversion ⇒
$h_{12} = (11|12) = (22|21) = 0$



The 2×2 CI problem

the OSS ($^1\Sigma_u^+$) is alone; the two closed shells ($^1\Sigma_g^+$) mix:

$$\mathbf{H} = \begin{pmatrix} E_1 & K_{12} \\ K_{12} & E_2 \end{pmatrix}, \qquad E_1 = 2h_{11} + J_{11}\ \text{(RHF)}, \quad E_2 = 2h_{22} + J_{22}$$

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also: $E(^3\Sigma_u^+) = h_{11} + h_{22} + J_{12} - K_{12}$, $\quad E(^1\Sigma_u^+) = h_{11} + h_{22} + J_{12} + K_{12}$

Pulling the atoms apart (1/2): the integrals

Assumptions

$R \to \infty$: $S \to 0$, all two-centre overlap densities $\chi_A \chi_B \to 0$; $U = (AA|AA)$, $V = (AA|BB) \to 1/R$

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Result

RHF misses the dissociation limit by $U/2$; one extra configuration fixes it exactly.

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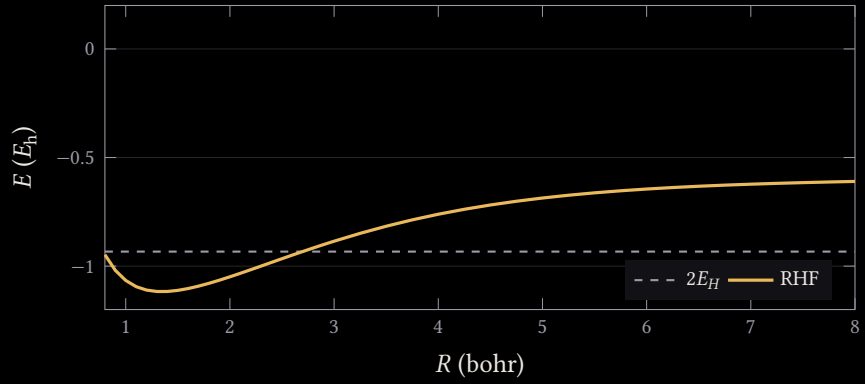
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RHF: 50% ionic
H^+H^- at infinity costs $U/2$

CI: purely covalent
Heitler–London H· ·H

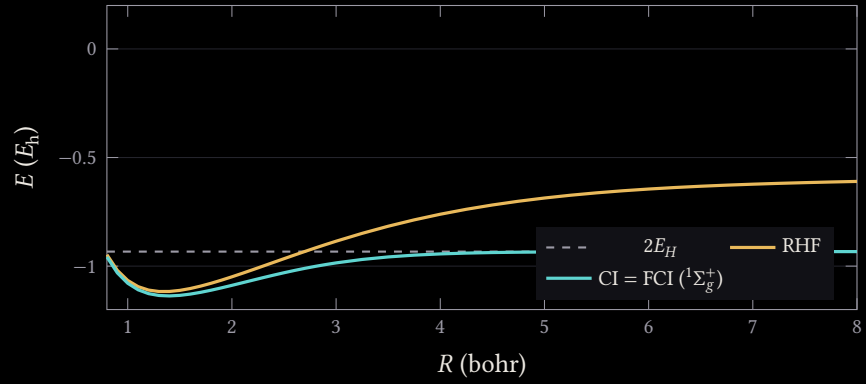
H_2 curves from our calculation

STO-3G, PySCF; closed forms agree to $4 \times 10^{-16}\ E_\mathrm{h}$ *(T5)*



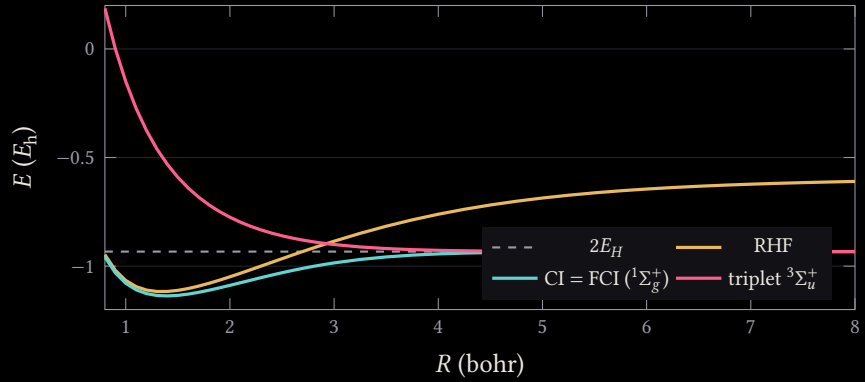
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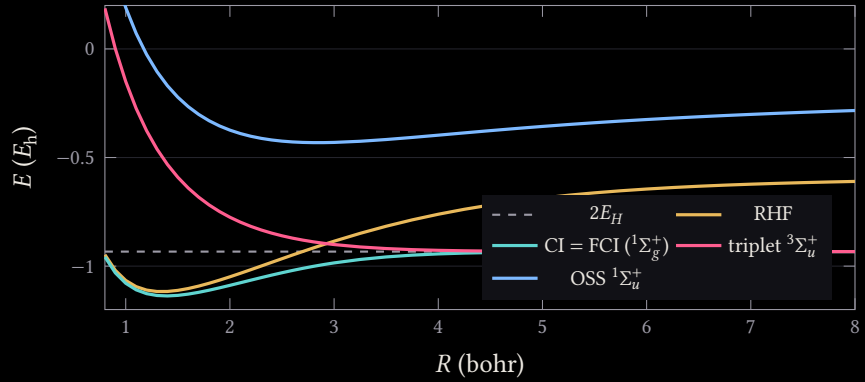
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Fig. 2b, twisted ethylene: the π/π^* gap $\Delta \to 0$ at 90°

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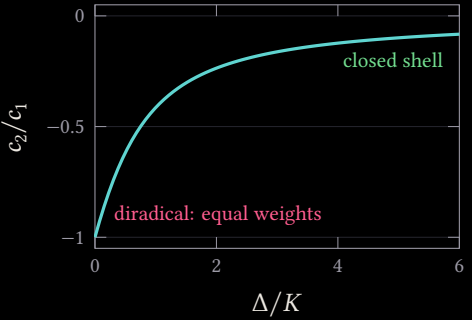
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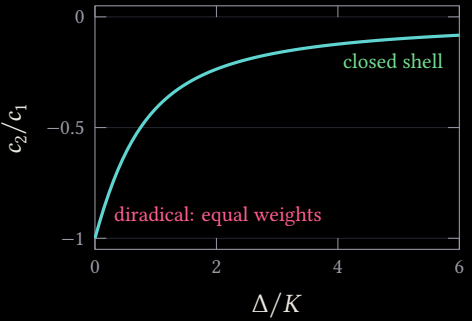
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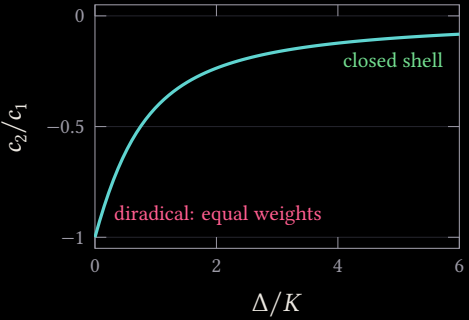
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VERIFIED weak interaction ⇒ strong static correlation (T6)

Part IX

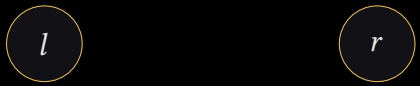
Kinetic exchange

IX

The Hubbard picture

The paper says

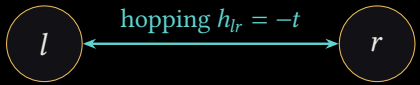
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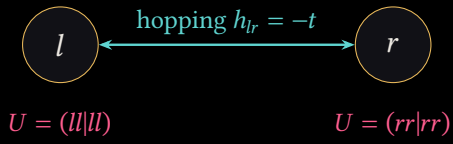
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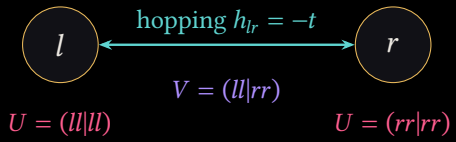
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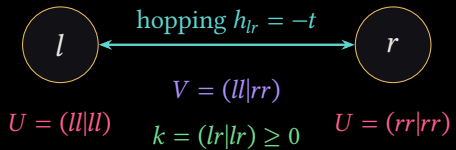
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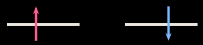
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Step 1: ionic configurations are pure singlets

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covalent



ionic l^2

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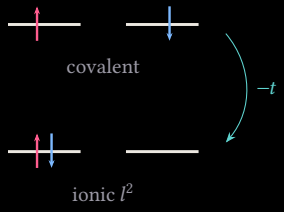
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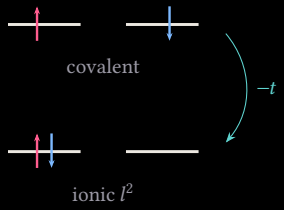
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Result

only the singlet can borrow from ionic configurations

Step 2: the matrix

insert the parameters into the $2e/2o$ Hamiltonian ($h_{ll} = h_{rr} = 0$, $(ll|lr) = 0$):

$$E_T = V - k, \qquad \mathbf{H}_S = \begin{pmatrix} U & k & -\sqrt{2}t \\ k & U & -\sqrt{2}t \\ -\sqrt{2}t & -\sqrt{2}t & V + k \end{pmatrix} \quad (ll,\ rr,\ \mathrm{cov})$$

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Φ_- decouples; the ground singlet lives in $\{\mathrm{cov}, \Phi_+\}$

Step 4: solve and expand

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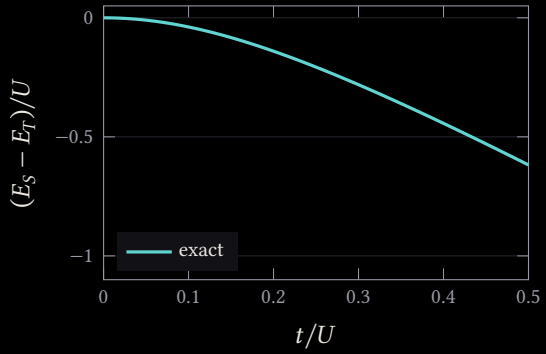
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$+2k$: potential exchange, favours the triplet

$-4t^2/(U-V)$: kinetic exchange, favours the singlet

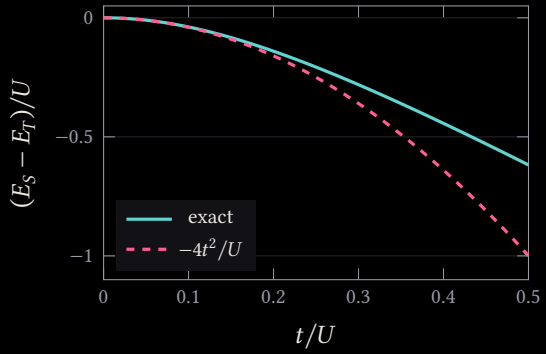
Hubbard dimer: exact versus kinetic exchange

$V = k = 0$: $E_S = \frac{1}{2}\left(U - \sqrt{U^2 + 16t^2}\right)$, $E_T = 0$



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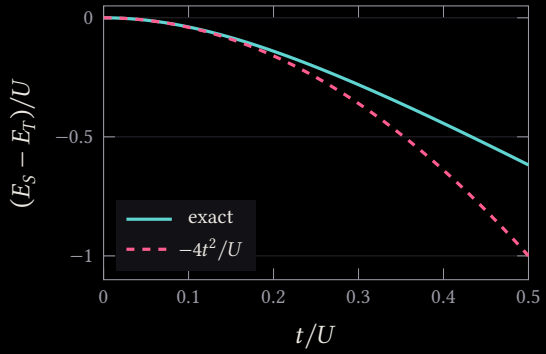
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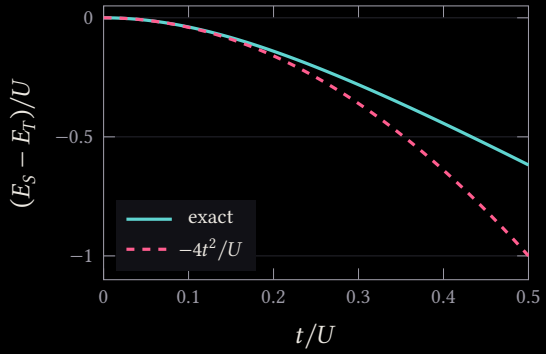
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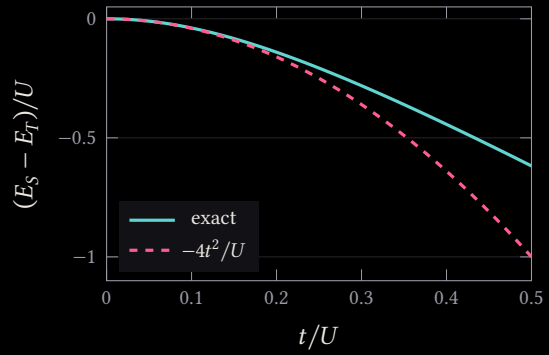
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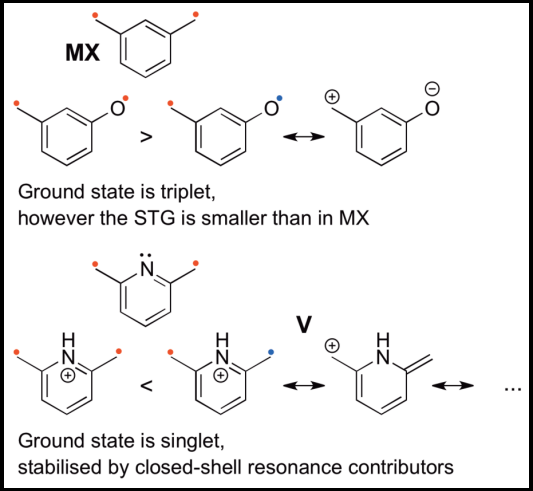
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✓ T7: symbolic + PySCF FCI (t=1, U=6, V=1.5, k=0.2)

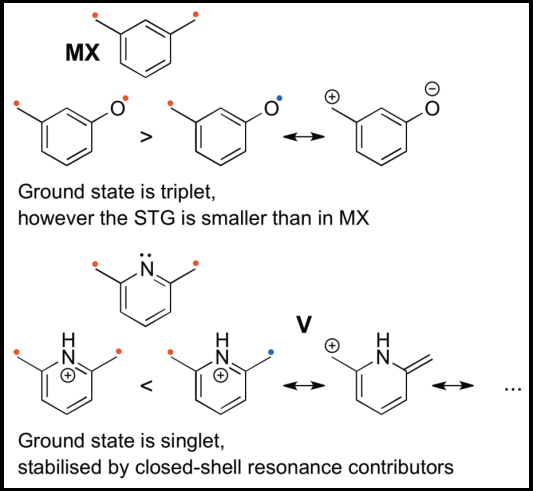
Ionic resonance reverses a spin state (Figure 6)



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"In the pyridinium analog of MX (V in Figure 6) this contribution exceeds the exchange and Coulomb terms, which formally favor the triplet state of this system with nondisjoint NBMOs."

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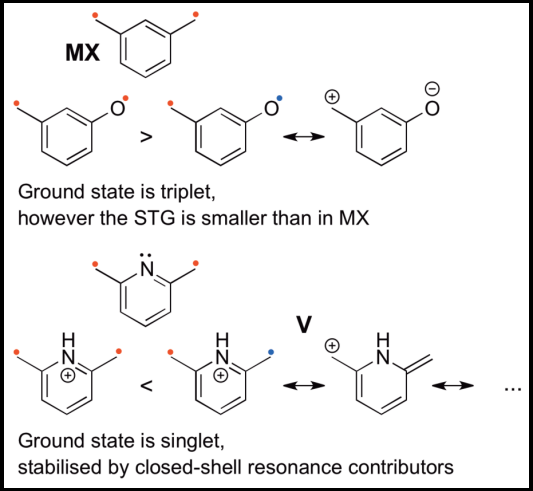
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(a qualitative mapping: the review gives no numbers)

Delocalised and localised: two views of one state

$$a, b = \frac{1}{\sqrt{2}}(l \pm r) \quad \Rightarrow \quad a(1)a(2) - b(1)b(2) = l(1)r(2) + r(1)l(2)$$

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same physics, different bookkeeping; the paper’s two pictures are compatible

Part X

NBMOs & disjointness



Borden & Davidson's classification

The paper says

"If the two orbitals do not have common atoms and the probability of two electrons occupying the same space is nil independent of their spins, they are termed disjoint. Under such circumstances, the energy of an open-shell singlet can become comparable to or even lower than that of a triplet." (p. 443)

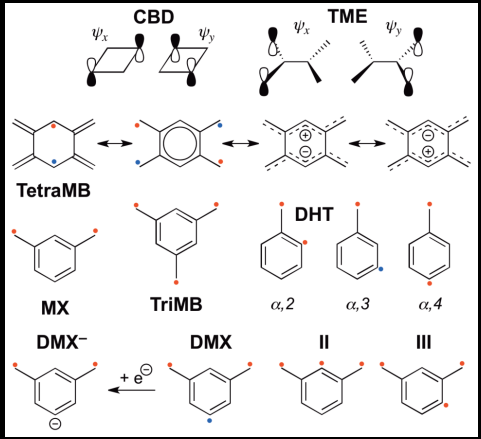


Fig. 3: high- and low-spin polyradicals

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three things to derive: (1) how to find NBMOs, (2) when they can be disjoint, (3) why disjoint means $K \approx 0$

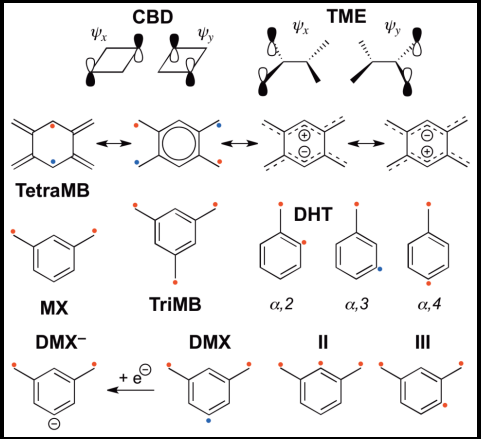


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Hückel theory and the zero-sum rule

Hückel

one p_π per carbon; $H_{ii} = \alpha$, $H_{ij} = \beta$ if bonded, else 0; zero overlap

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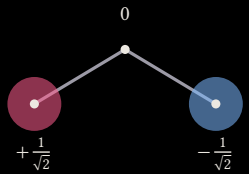
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allyl NBMO: centre sees $c_1 + c_3 = 0$

Alternant hydrocarbons: the pairing theorem

star every other atom; bonds only join starred ($\mathscr{A}$) to unstarred ($\mathscr{B}$):

$$\mathbf{A} = \begin{pmatrix} \mathbf{0} & \mathbf{B} \\ \mathbf{B}^{\mathsf{T}} & \mathbf{0} \end{pmatrix}$$

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levels come in $\pm x$ pairs, mirrored about α (Lean: `pairing`)

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Careful: only a bound

square cyclobutadiene: $N = 4$, $T = 2$, so $N - 2T = 0$, yet it has *two* NBMOs



perfect matching

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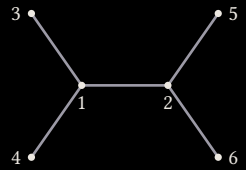
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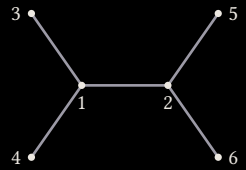
T8 caught our own wrong assumption here.

Tetramethyleneethane: solving the zero-sum equations



atom 3: $c_1 = 0$

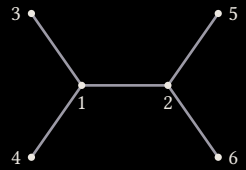
Tetramethyleneethane: solving the zero-sum equations



atom 3: $c_1 = 0$

atom 5: $c_2 = 0$

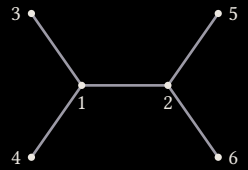
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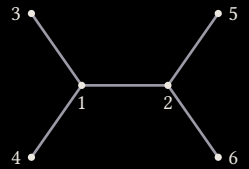


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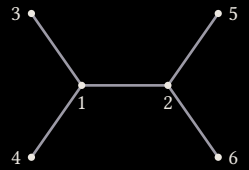
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atom 1: $c_2 + c_3 + c_4 = 0 \Rightarrow c_4 = -c_3$

atom 2: $c_1 + c_5 + c_6 = 0 \Rightarrow c_6 = -c_5$

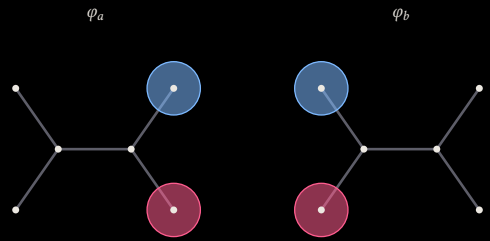
$$\varphi_a = \frac{1}{\sqrt{2}}(\chi_3 - \chi_4), \qquad \varphi_b = \frac{1}{\sqrt{2}}(\chi_5 - \chi_6)$$

no common atom: disjoint

Disjoint pairs, drawn by our JavaScript

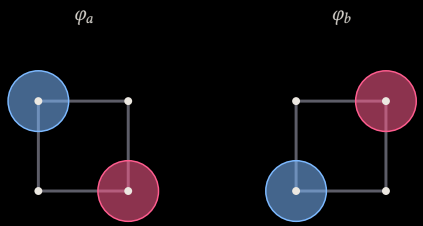
lobe area $\propto |c_i|^2$; *rose* $+$, *blue* $-$

tetramethyleneethane



disjoint: $\Sigma\, a_i^2\, b_i^2 = 0 \Rightarrow K_{ab} = 0$

cyclobutadiene (D_4h)



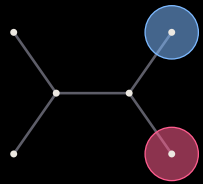
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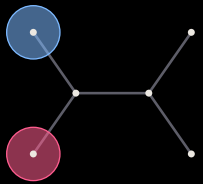
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φ_a



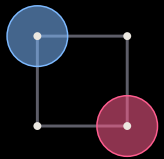
φ_b



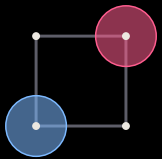
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cyclobutadiene (D_4h)

φ_a



φ_b



disjoint: $\Sigma\, a_i^2\, b_i^2 = 0 \Rightarrow K_{ab} = 0$

JS code: Hückel matrix → Jacobi diagonalisation → rotate the degenerate pair to minimise $\sum_i a_i^2 b_i^2$ → SVG → PDF

Trimethylenemethane cannot be made disjoint

1. zero-sum at a terminal $\Rightarrow c_0 = 0$

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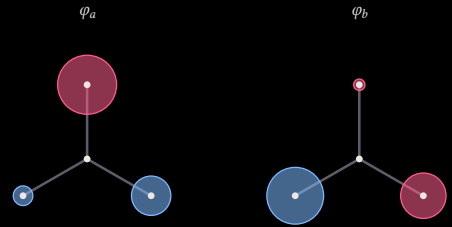
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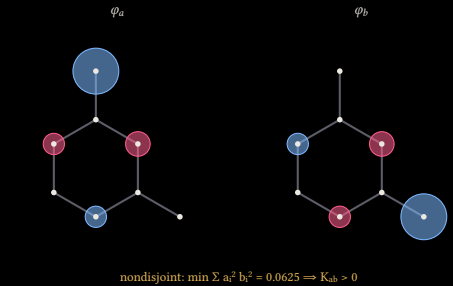
nondisjoint: min $\Sigma\, a_i^2\, b_i^2 = 0.1667 \Rightarrow K_{ab} > 0$

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nondisjoint, for every choice of basis

m-xylylene



nondisjoint: min $\Sigma\, a_i^2\, b_i^2 = 0.0625 \Rightarrow K_{ab} > 0$

m-xylylene: also nondisjoint

Why disjoint means $K_{ab} = 0$

Zero differential overlap (ZDO)

$(\mu\nu|\lambda\sigma) = \delta_{\mu\nu}\delta_{\lambda\sigma}\,\gamma_{\mu\lambda}$, γ = Coulomb repulsion between atoms μ, λ

$$K_{ab} = (ab|ab) = \sum_{\mu\nu\lambda\sigma} a_\mu b_\nu a_\lambda b_\sigma \,(\mu\nu|\lambda\sigma)$$

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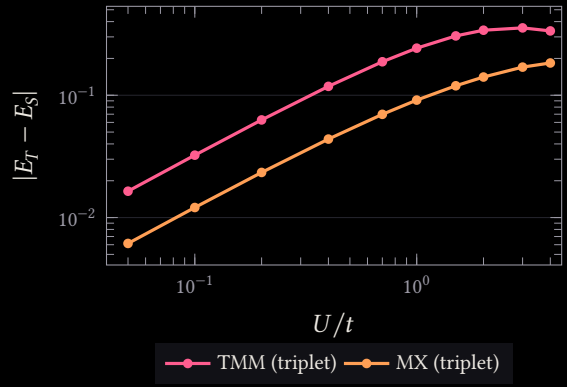
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TMM: $\min \sum a^2 b^2 = \frac{1}{6}$ MX: $\frac{1}{16}$ (Python *and* JavaScript agree)

The test: how the gap grows with U

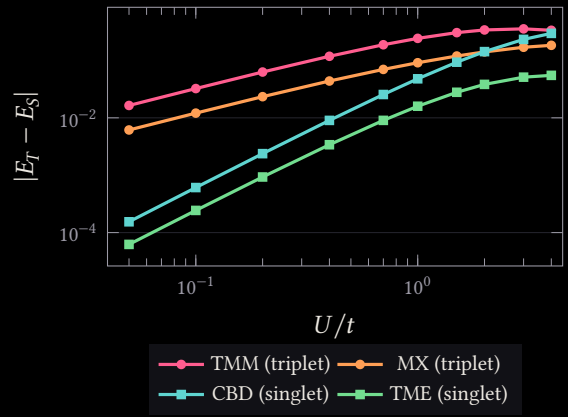
T9b: exact Hubbard diagonalisation, $t = 1$



- nondisjoint: slope ≈ 1; at $U = 0.05$: TMM 0.0164 vs $2U \cdot \frac{1}{6} = 0.0167$

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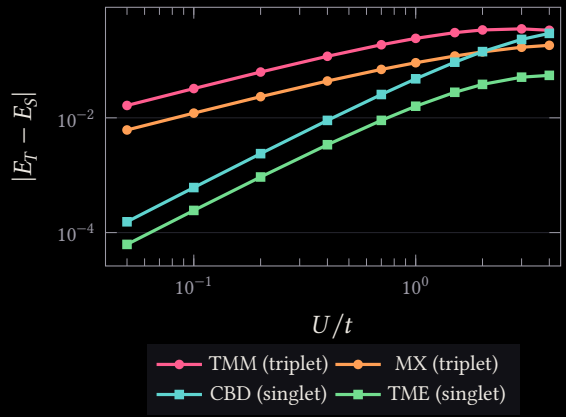
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- disjoint: slope ≈ 2 (fit 1.97, 1.95), singlet below triplet
- first-order K vanishes; something second-order picks the singlet…

Part XI

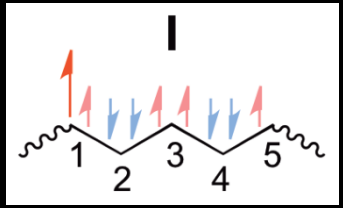
Spin polarisation

XI

Dynamic spin polarisation

The paper says

"A damped spin wave so formed generates ferromagnetic coupling between the spins separated by an odd number of sites, and antiferromagnetic between those separated by an even number of sites, thus yielding the topology rules of spin coupling." (p. 443)

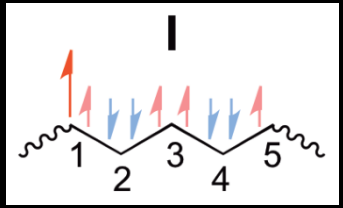


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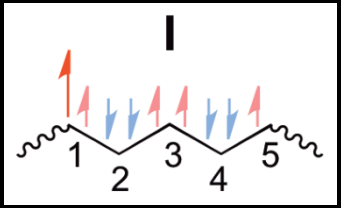
Scheme 1

1. Hückel: the unpaired electron sits in the NBMO, which is zero on unstarred atoms

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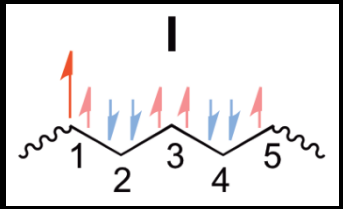
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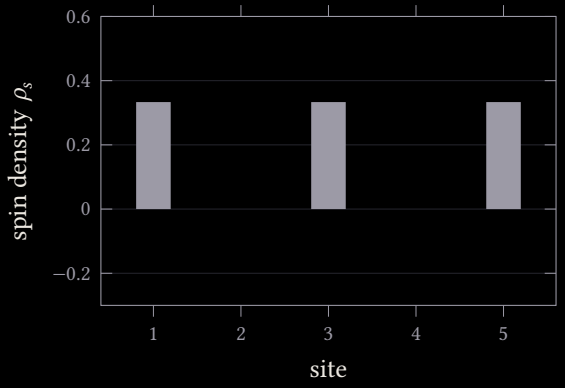


Scheme 1

1. Hückel: the unpaired electron sits in the NBMO, which is zero on unstarred atoms
2. add on-site repulsion U: an ↑ electron pushes ↓ density onto its neighbours
3. negative spin density appears between positive ones: a correlation effect

Measured on the pentadienyl radical

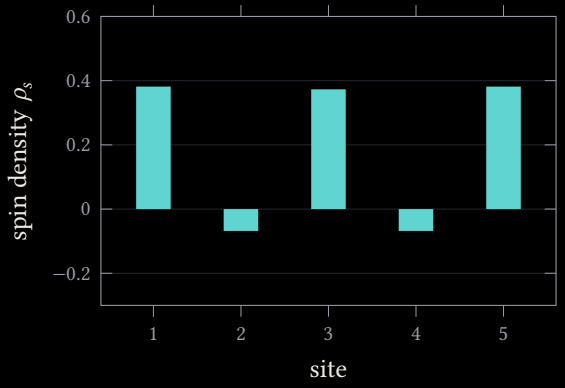
T9: exact Hubbard diagonalisation, 5 sites, 5 electrons, $S_z = \frac{1}{2}$



$U = 0$ (Hückel):
unstarred sites exactly 0

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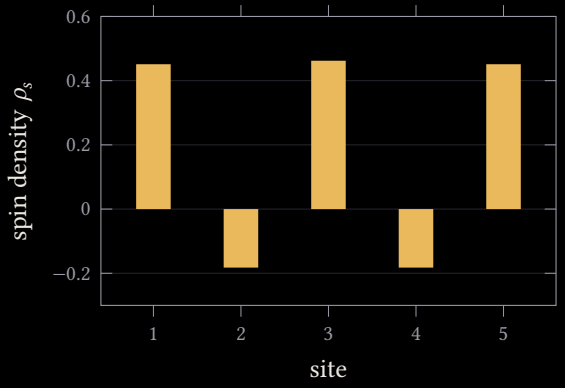
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$U = t$: small negative dips

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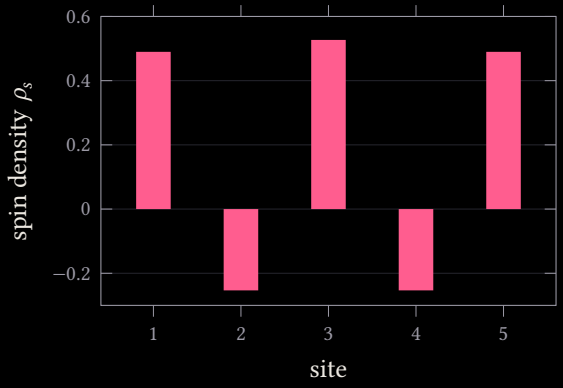
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$U = 3t$: -0.18 on sites 2, 4

Measured on the pentadienyl radical

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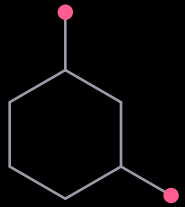


$U = 6t$: -0.25; the “damped spin wave”

total is always 1; positive on starred, negative on unstarred atoms

From bonds to topology

1. strong U: each bond carries $J_{\mathrm{AF}} = 4t^2/U$ (kinetic exchange)

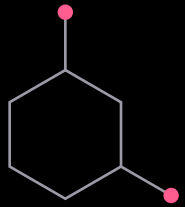


meta: 3 atoms between

parallel, triplet

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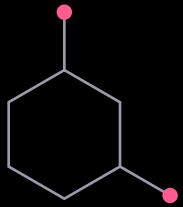


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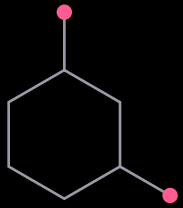
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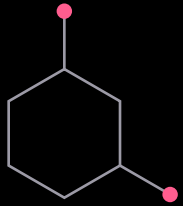
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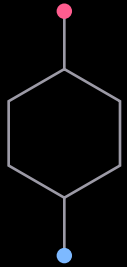
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same set: parallel; different sets: antiparallel



meta: 3 atoms between

parallel, triplet



para: 4 atoms between

antiparallel, singlet

Ovchinnikov's rule, and Lieb's theorem behind it

count the Néel arrangement: n_A up, n_B down

$$S = \frac{1}{2} |n_A - n_B|$$

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Lieb, Phys. Rev. Lett. 62, 1201 (1989)

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T9 ($U = 3t$)	n_A/n_B	rule	exact
CBD	2/2	0	0
TMM	1/3	1	1
TME	3/3	0	0
MX	3/5	1	1
TriMB	3/6	$\frac{3}{2}$	$\frac{3}{2}$
TetraMB	5/5	0	0
p-xylylene	4/4	0	0

Disjoint diradicals: who picks the singlet?

The paper says

"In the singlet state of D_{4h} CBD, the bonding π-electrons can partially localize on the same pair of atoms as the same-spin unpaired electrons; however, in the triplet state such advantageous spin polarization is not possible." (p. 443)

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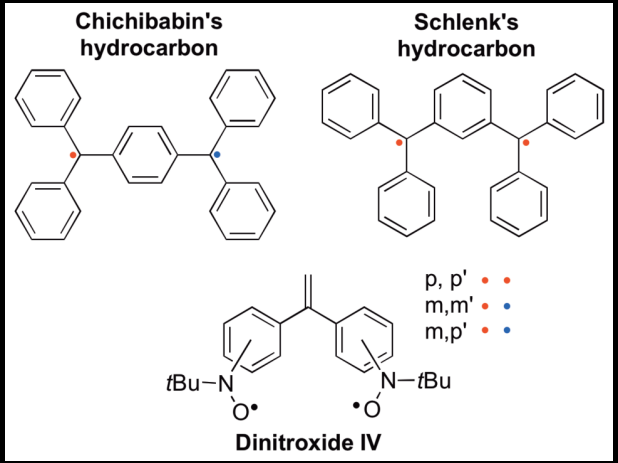
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VERIFIED consistent with Kollmar & Staemmler's mechanism
(their paper itself was paywalled)

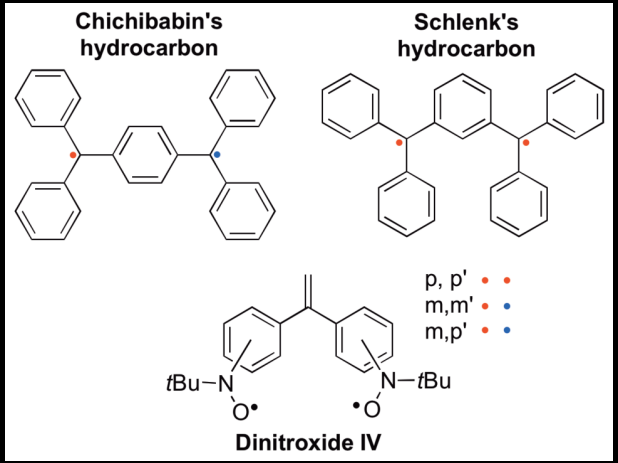
Topology in the paper: Figure 4



The paper says

simple polyradicals "can also be thought of as coupling units that sustain certain spin alignment depending on topology, as seen in Schlenk's and Chichibabin's hydrocarbons, or in isomers of dinitroxide IV"

Topology in the paper: Figure 4

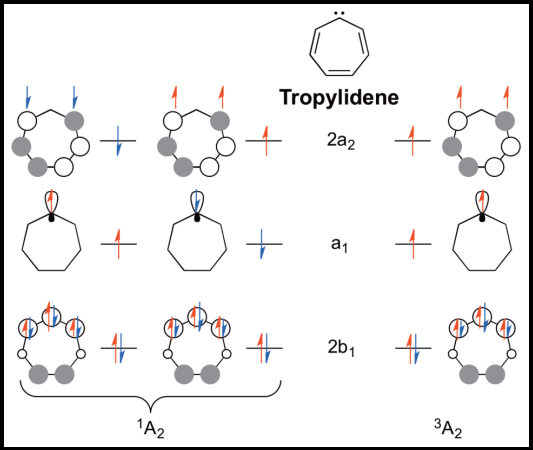


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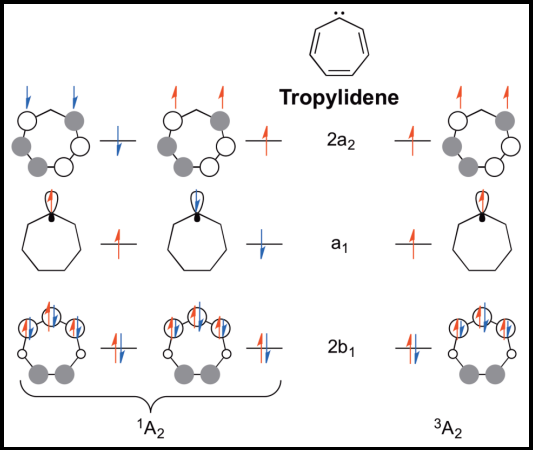
p, p' and m, m' / m, p' isomers of IV differ only in topology, hence in the coupling sign

Tropylidene: a singlet from spin polarisation (Figure 5)



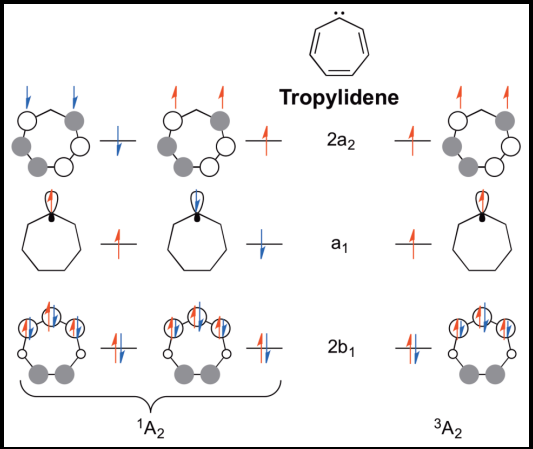
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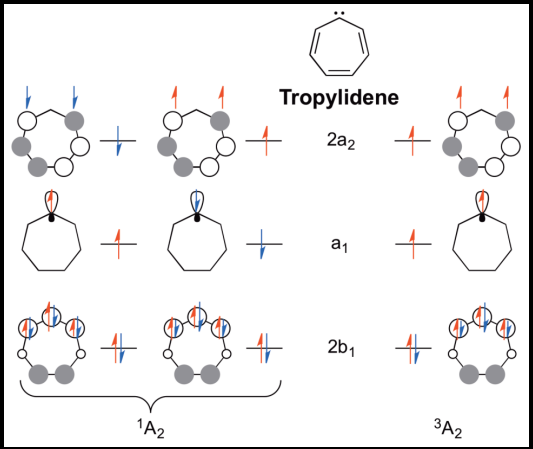
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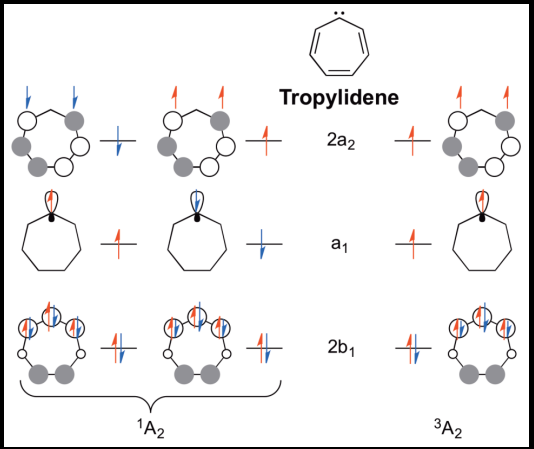
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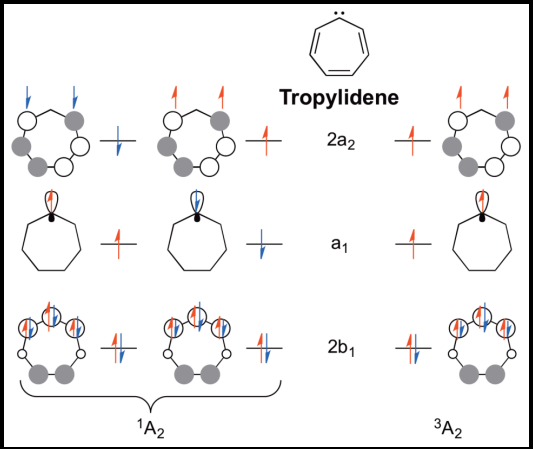
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Hund's rule obeyed locally, violated globally.

Where the rules stop

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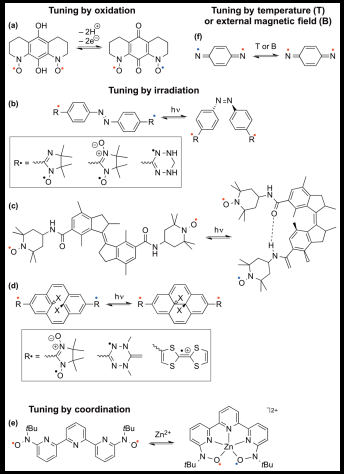
VERIFIED fair: each rule above *is* a theorem of a simplified model

Part XII

Switching spin states

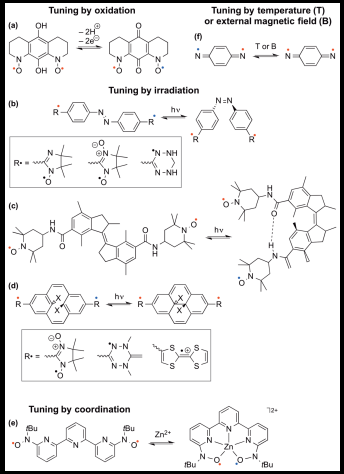
XII

Spin-state crossover by external stimuli (Figure 7)



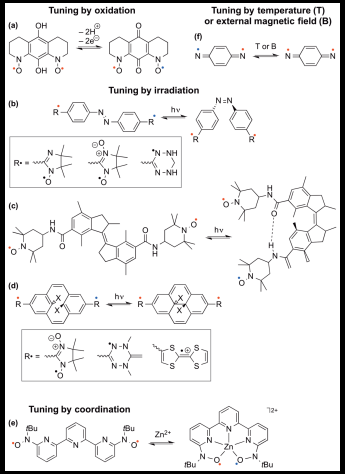
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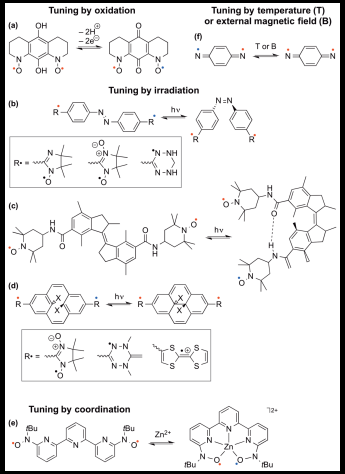
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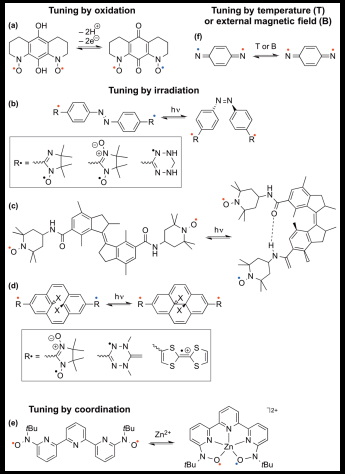
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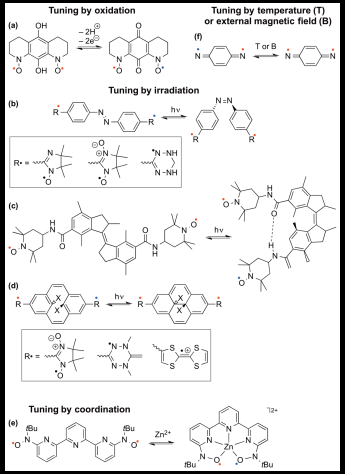
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- (f) *T* or *B*: 1,4-phenylenedinitrene, open-shell singlet → triplet

A wording slip in (e)?

The paper says

> "Coordination to Zn^{2+} brings about O–Zn–O superexchange and O···O through-space exchange, which favor antiferromagnetic coupling and override the preference for the triplet ground state, predicted on the basis of *disjoint* NBMOs in parent dinitroxide ligand" (p. 445)

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OVERSTATED inconsistent wording; original (ref. 52) not checked

Panel (f), quantitatively: temperature

Assumptions

open-shell singlet ground state, triplet Δ above it; Boltzmann statistics; degeneracies 1 and 3

$$Z = 1 + 3e^{-\Delta/k_B T}$$

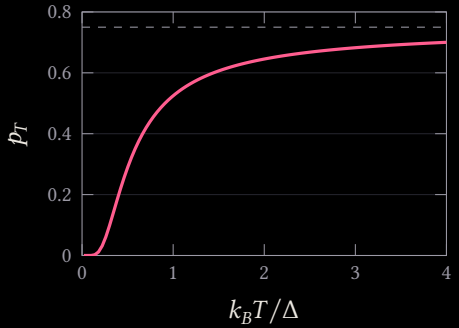
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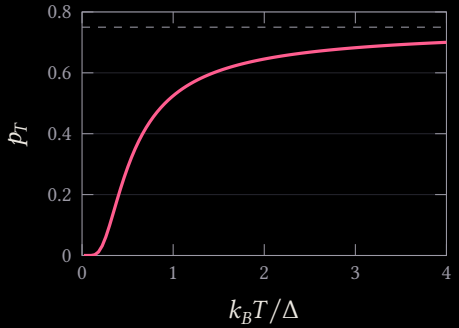
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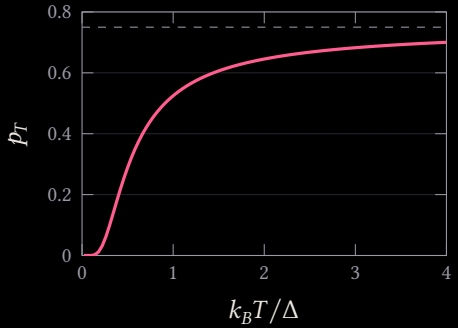
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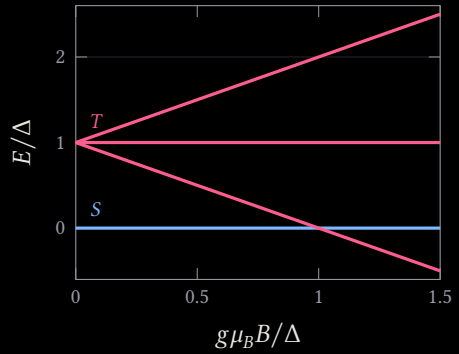
"increased temperature … promotes electrons to populate the triplet state" VERIFIED

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Zeeman term $g\mu_B B M_S$; singlet unaffected; $g \approx 2$

$$E_T(M_S) = \Delta + g\mu_B B\, M_S, \qquad M_S = 0, \pm 1$$



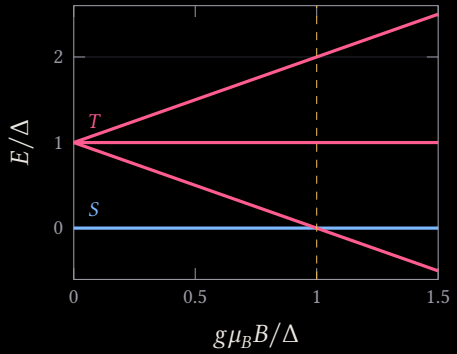
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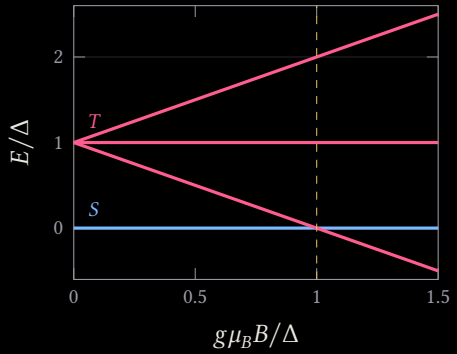
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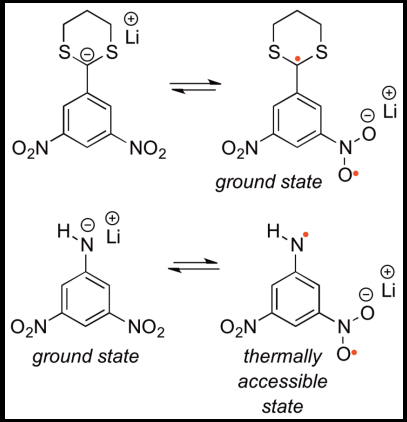
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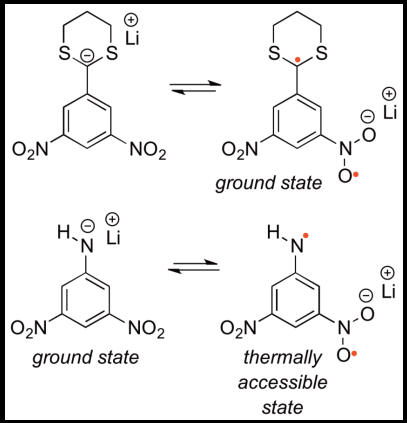
"external magnetic field that splits the triplet into microstates, one of which approaches the energy of the ground-state singlet" VERIFIED

Closed shell or diradical? (Figures 8 and 9)



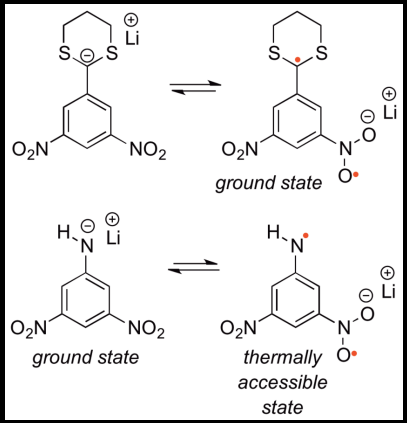
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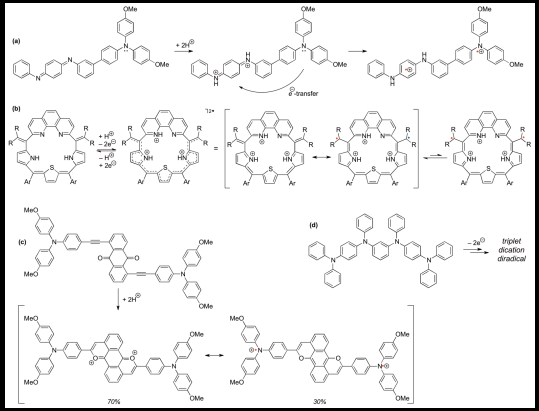
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- singlet fission: one excited singlet shares its excitation with a neighbour; both end in a low triplet

Protonation-induced magnetisation (Figure 9)



descriptive part of the review: no mathematics to derive, only claims about specific molecules (not re-checked)

Part XIII

Koopmans & Janak

XIII

The claim

The paper says

"An alternative theoretical interpretation of the Hund's multiplicity rule has been proposed on the basis of Janak theorem (or extended Koopmans' theorem) …: at the Hartree–Fock level, the singlet-triplet gap (STG) is equal to the difference between the energies of the triplet's and open-shell singlet's HOMOs." (p. 442–443)

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(which difference, which sign? the review does not say; the derivation will)

Koopmans' theorem

Assumptions

single determinant of spin orbitals χ_i; remove χ_k *without* relaxing the others

$$E_N = \sum_i \langle i|\hat{h}|i\rangle + \frac{1}{2}\sum_{i,j}\big([ii|jj] - [ij|ji]\big)$$

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\mathrm{IP}_k = E_{N-1}^{(k)} - E_N = -\varepsilon_k \qquad \text{(frozen orbitals)}
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Janak's theorem

Kohn--Sham with fractional occupations n_i

$$E = \sum_i n_i \langle \psi_i | -\frac{1}{2}\nabla^2 | \psi_i \rangle + \int v\rho + E_H[\rho] + E_{xc}[\rho], \qquad \rho = \sum_i n_i |\psi_i|^2$$

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$\boxed{\partial E / \partial n_i = \varepsilon_i}$ for any E_{xc} (Janak, Phys. Rev. B 18, 7165, 1978)

The gap as a HOMO difference

define each state's "HOMO energy" as minus the cost of removing the φ_b electron, φ_a frozen; ion: $E_{\text{ion}} = h_a$

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Result

true at frozen orbitals; the triplet's HOMO is lower (more bound, "harder") by $2K$. Like Slater's argument it has no relaxation and no dynamic correlation, as the review notes.

VERIFIED

Part XIV

Aufbau on trial

XIV

The prosecution

The paper says

"the aufbau principle, by assuming the existence of unchanging stationary individual electronic states, violates the Pauli exclusion principle. Only the many-electron system as a whole possesses stationary states." (p. 448)

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the adiabatic argument behind "building up"

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one-electron quantum numbers are not conserved

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Exhibit C
the 3d/4s paradox

Exhibit A: $[\hat{l}_{1z}, 1/r_{12}] \neq 0$

$$\hat{l}_{1z} = -i(x_1\partial_{y_1} - y_1\partial_{x_1}), \quad \partial_{y_1} r_{12}^{-1} = -\frac{y_1 - y_2}{r_{12}^3}, \quad \partial_{x_1} r_{12}^{-1} = -\frac{x_1 - x_2}{r_{12}^3}$$

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\end{aligned}$$

first-order operator: $[\hat{l}_{1z}, V]f = (\hat{l}_{1z}V)f \quad \Rightarrow \quad l_{1z}$ is not a constant of motion

. . . but the total is conserved

$$\hat{l}_{2z}\, r_{12}^{-1} = -i\,\frac{x_2 y_1 - x_1 y_2}{r_{12}^3} \quad (\partial_{y_2} r_{12}^{-1} = +\tfrac{y_1 - y_2}{r_{12}^3})$$

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$[\hat{H}, \hat{\mathbf{L}}] = 0,\ [\hat{H}, \hat{S}^2] = 0,$
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$\hat{\mathbf{L}}$ rotates both electrons together (keeps r_{12}); $\hat{\mathbf{l}}_1$ rotates one (changes r_{12}).
✓ T13, SymPy on an arbitrary f

Exhibit B: the adiabatic theorem

The paper says

> Bohr assumed stationary states survive “addition of an electron and a proton”; this rests on Ehrenfest’s adiabatic principle, which “does not apply to aperiodic systems” (p. 448)

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slow $\dot{\hat{H}}$, finite gaps ⇒ the system stays in the *continuously connected* eigenstate. But: adding a particle changes the Hilbert space (no continuous path), and the conserved labels are L, S of the *whole* system, never one-electron labels.

Exhibit C: the 3d/4s paradox

The paper says

4s lies slightly above 3d when both are populated, so 4s is ionised first; “however, 4s orbital is also populated first, because it offers less repulsion with a 3d electron than that between two 3d electrons” (p. 448)

Average of configuration

one-electron energies h_d, h_s; one average repulsion per pair type: $\bar{J}_{dd}$, $\bar{J}_{ds}$, J_{ss}

$$E(d^{n_d}s^{n_s}) = n_d h_d + n_s h_s + \binom{n_d}{2}\bar{J}_{dd} + n_d n_s \bar{J}_{ds} + \binom{n_s}{2} J_{ss}$$

3d/4s: the algebra

$$E(d^n s^2) = n h_d + 2h_s + \binom{n}{2} \bar{J}_{dd} + 2n \bar{J}_{ds} + J_{ss}$$

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$$\Delta \equiv E(d^{n+1}s) - E(d^n s^2) = h_d - h_s + n\bar{J}_{dd} + (1-n)\bar{J}_{ds} - J_{ss} \quad \binom{n+1}{2} - \binom{n}{2} = n$$

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Result

$$\Delta = (\varepsilon_d - \varepsilon_s) + \left(\bar{J}_{dd} - \bar{J}_{ds}\right) \qquad \text{(T12; Lean } \texttt{three_d_four_s}\text{)}$$

3d/4s: reading the identity

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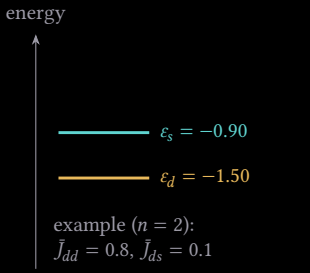
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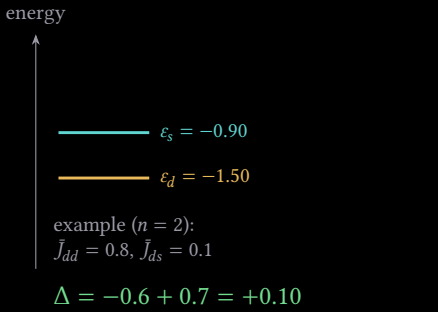
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orbital energies belong to a *configuration*: the building-up and removal orders need not agree

Verdict on the aufbau critique

right

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- orbital energies depend on the configuration (Exhibit C)

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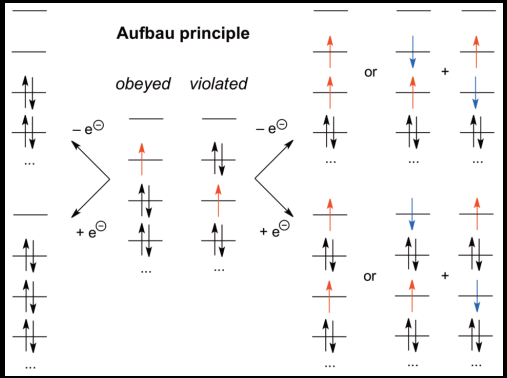
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Part XV

Quasi-closed shells

XV

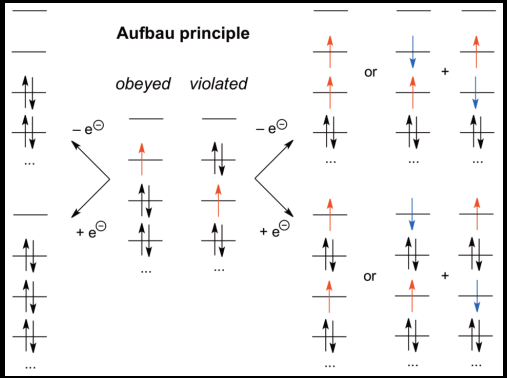
SOMO below HOMO (Figure 10)



The paper says

"upon ionization an electron is removed from the higher energy doubly occupied MO, leaving another unpaired electron behind, while upon addition new electron enters the former LUMO; essentially, SOMO does not participate in these processes." (p. 449)

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the fingerprint is redox: products are diradicals, not closed shells

UHF orbital energies of s^1h^2 (1/2)

Assumptions

frozen orbitals; SOMO s (α only), doubly occupied h; core absorbed into h_s, h_h; $\varepsilon_k = E_N - E_{N-1}^{(k)}$ (Koopmans)

configuration: $s\alpha,\ h\alpha,\ h\beta$ pairs: $(s\alpha, h\alpha)$ same spin, $(s\alpha, h\beta)$, $(h\alpha, h\beta)$

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$$\varepsilon_h^\beta = E_N - E(s\alpha, h\alpha) = h_h + J_{hh} + J_{hs}$$

UHF orbital energies of s^1h^2 (2/2)

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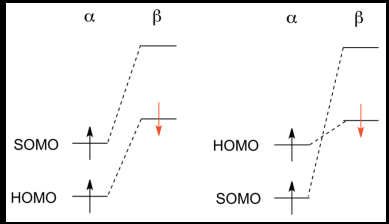
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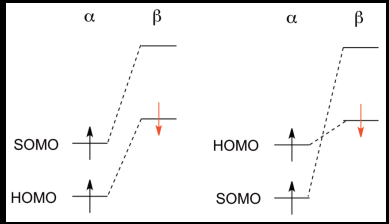
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✓ T2: all four checked against determinant energies

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The paper says

"spatially separated stable radical center and molecular fragment serving as a source of high-energy doubly occupied orbitals" (p. 454)

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the large on-site J_{ss} is what pushes ε_s^β up

ROHF: are these orbital energies even real?

The paper says

> "erroneous non-aufbau occupations can be observed with restricted open-shell methods depending on the canonicalization parameters of the α and β electrons in Fock matrix" (p. 449)

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a rotation changes $\mathbf{D}^\sigma$ only if exactly one of p, q is occupied in spin σ

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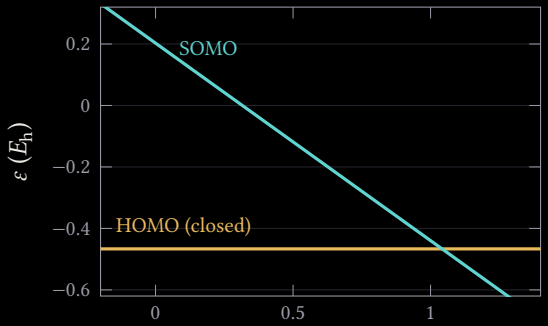
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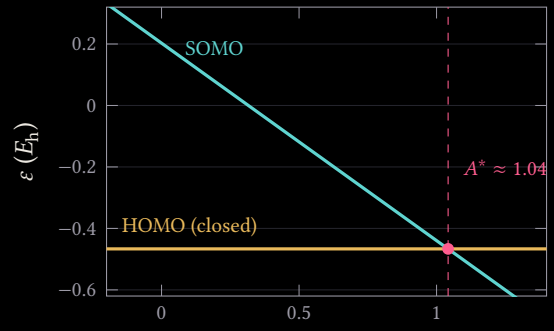
✓ T11a: required blocks $< 10^{-7}$, others ~ 0.1 (H_2NO, NO_2, CH_3, allyl); PySCF's Roothaan Fock uses exactly these off-diagonal blocks

How arbitrary? H_2NO, ROHF/6-31G*



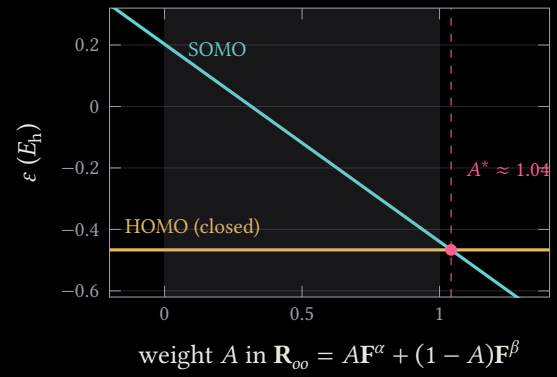
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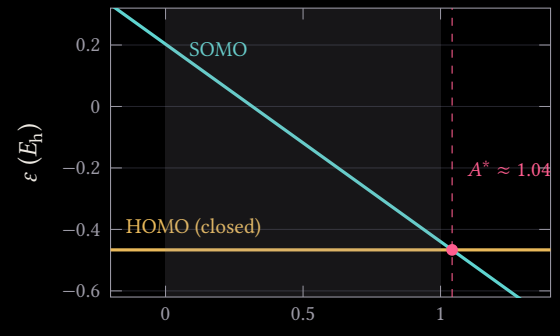
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OVERSTATED the review is right in principle; for ordinary radicals the standard choices do not flip

Hence: trust observables

The paper says

"a rigorous support for the non-aufbau occupation can be obtained a posteriori from the ionization potentials and electron affinities (or redox potentials in solution) … as well as from the determination of the spin state and magnetic properties of the corresponding open-shell oxidation/reduction products." (p. 449)

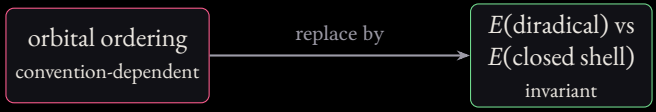
orbital ordering
convention-dependent

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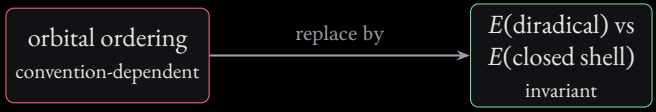
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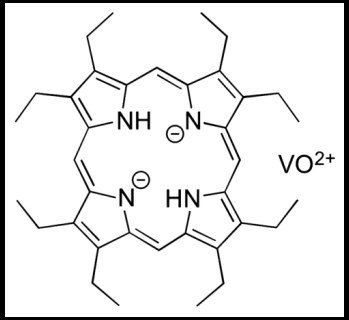
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this is exactly the test we ran on a distonic radical anion (a few slides ahead)

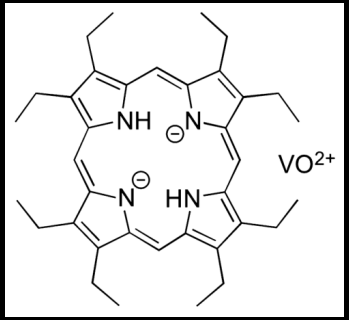
Inorganic precedents (Scheme 2)



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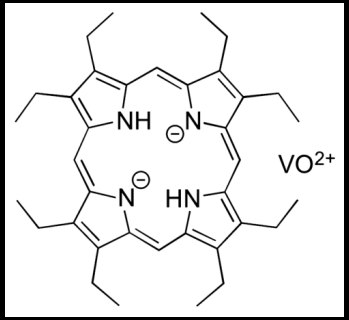
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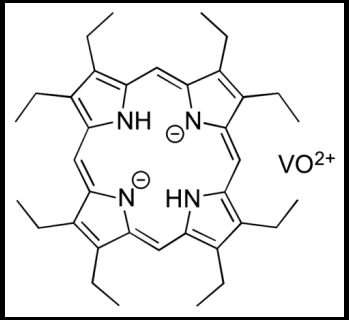
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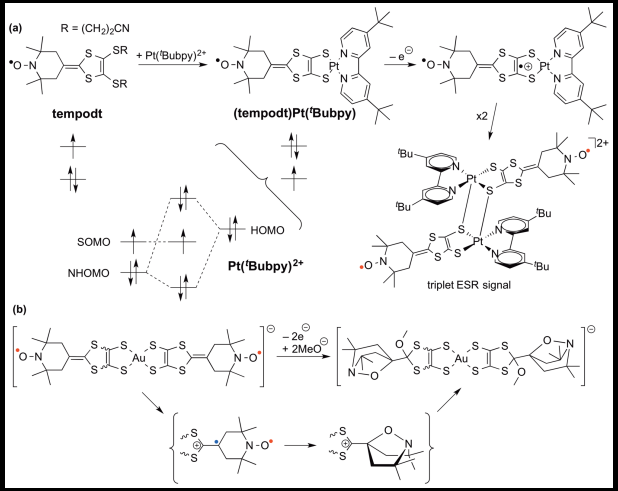


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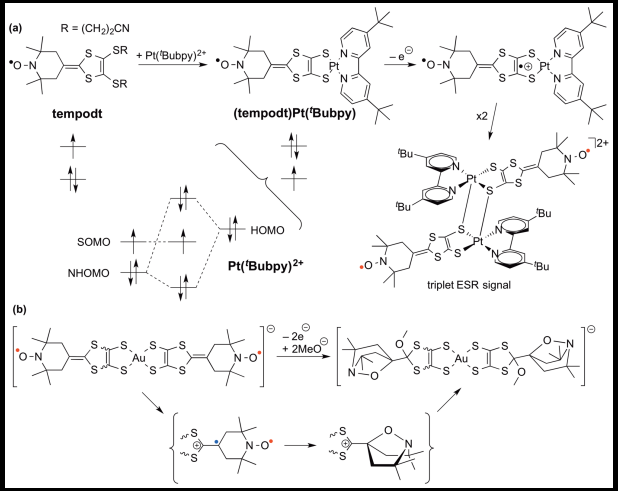
also reported: quantum dots, the LiF dissociation curve

A TEMPO-dithiolate on platinum (Figure 11)



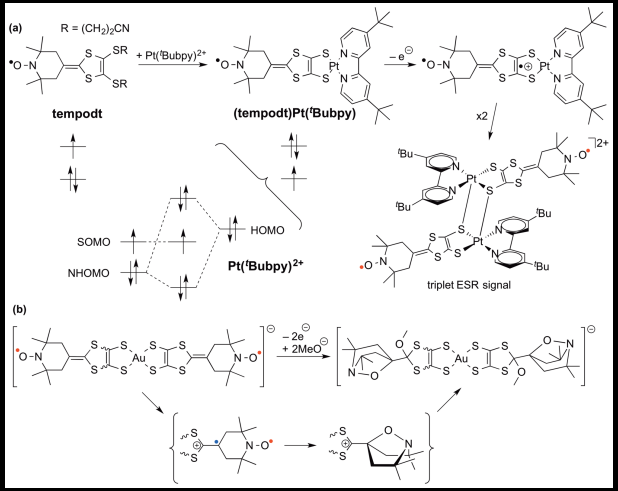
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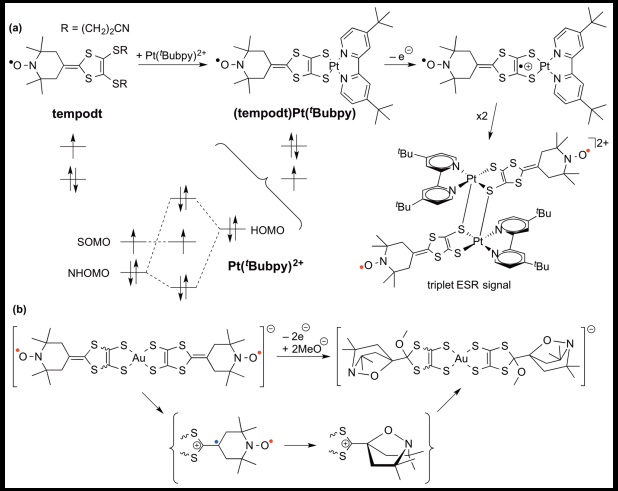
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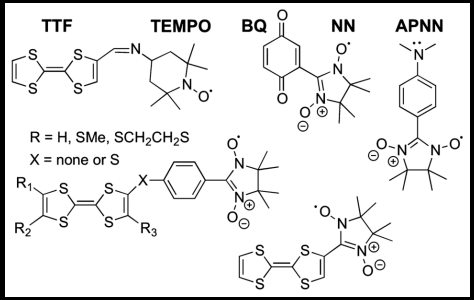
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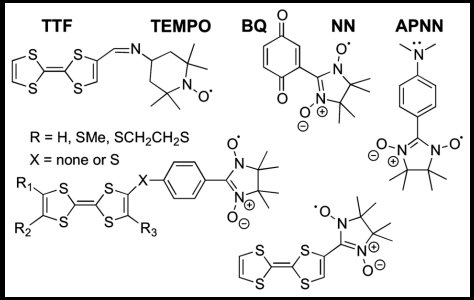
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Spin-polarised donors (Figures 12, 13)

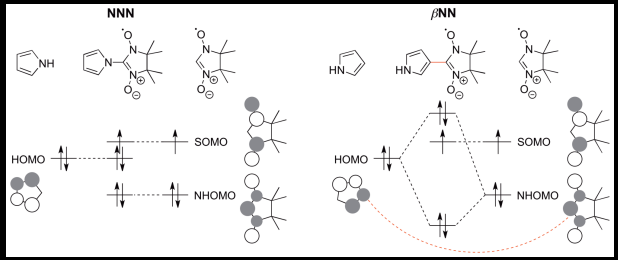


nitronyl nitroxide (NN) radicals on TTF, BQ, amine donors: one-electron oxidation gives triplet cation diradicals

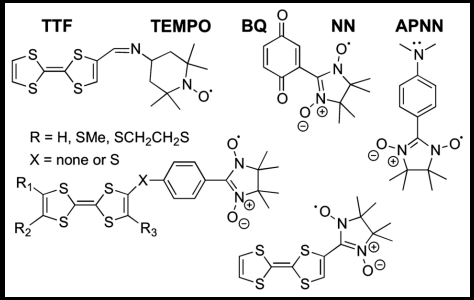
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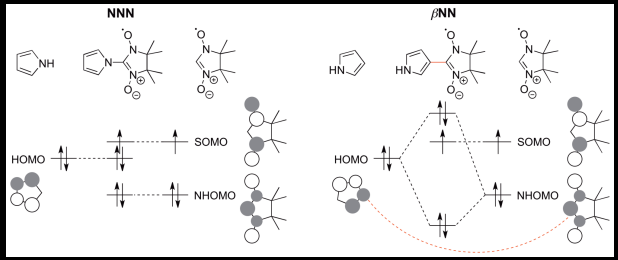
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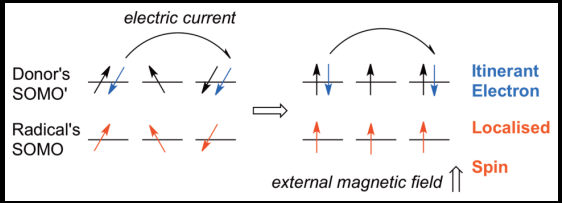


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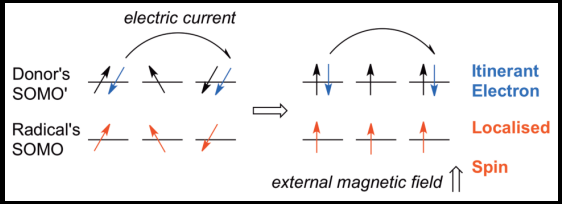
NNN (1-pyrrolyl): donor HOMO disjoint from SOMO/NHOMO. *β*NN (3-pyrrolyl): donor HOMO and NHOMO nondisjoint; their combination rises above the SOMO.

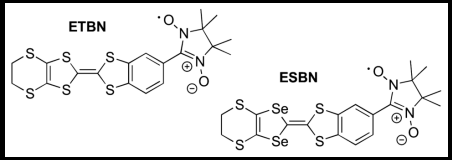
Conducting magnets (Scheme 4, Figure 14, Scheme 5)



- two nondegenerate SOMOs: the donor's (higher) carries an itinerant electron, the radical's (lower) a localised spin

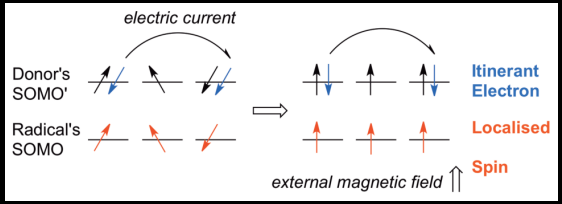
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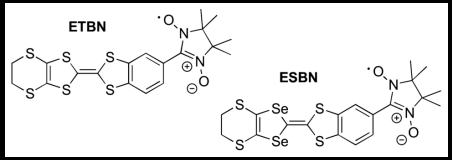




- two nondegenerate SOMOs: the donor's (higher) carries an itinerant electron, the radical's (lower) a localised spin
- ETBN, ESBN perchlorate salts: "first genuine paramagnetic conductors"

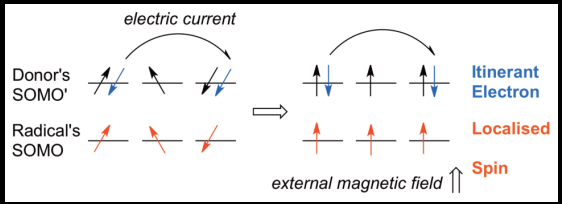
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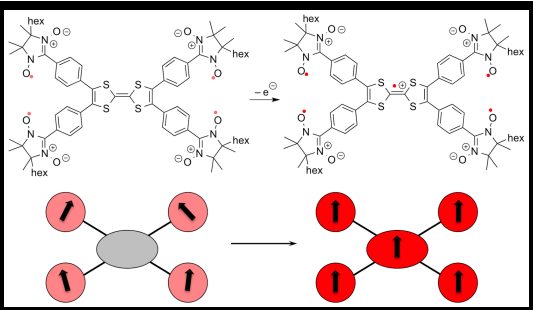




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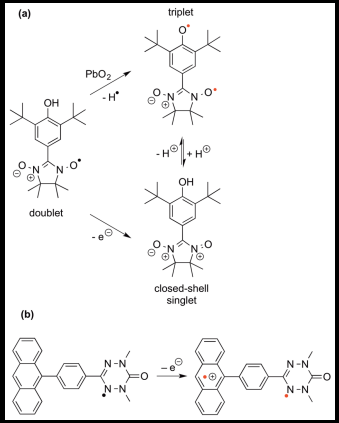
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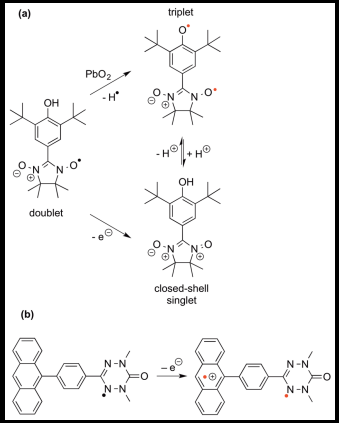
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- Fig. 14: several radicals on one donor; oxidation gives a strongly ferromagnetic high-spin system

Other radicals, and a pH sensor (Figure 15)



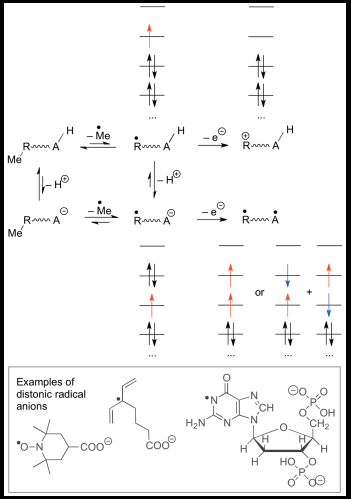
- (a) NN with a phenol-based donor: (de)protonation switches between *triplet* and *closed-shell singlet*: a pH sensor

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- (b) anthracene–*p*-phenylene–oxoverdazyl (AnOV): high-spin oxidation products

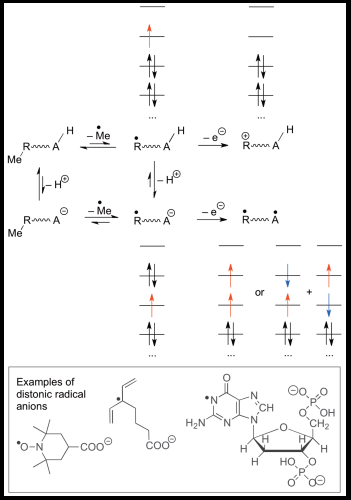
Distonic radical anions (Figure 16)



The paper says

a stabilised radical plus an anionic group not π-conjugated to it: the anion's HOMO lies above the SOMO; "if the anion is protonated, the aufbau orbital occupation is restored", switchable "via changing pH"; on oxidation the conversion "yield[s] diradical species preferentially over the closed-shell zwitterion" (p. 452–453)

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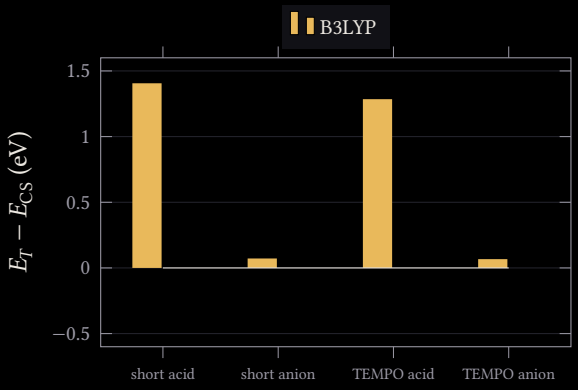
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testable: compute both oxidised states, with and without the proton

Our test: two radicals, two protonation states, two functionals

T11/T11c: UB3LYP/6-31G geometries, vertical oxidation at 6-31+G**

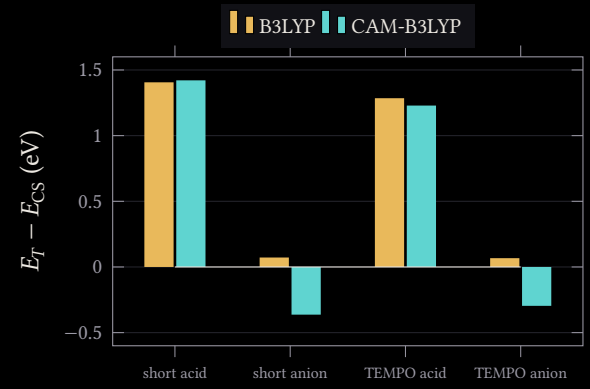


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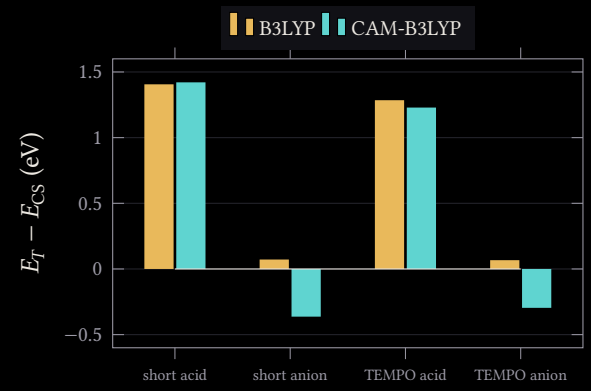


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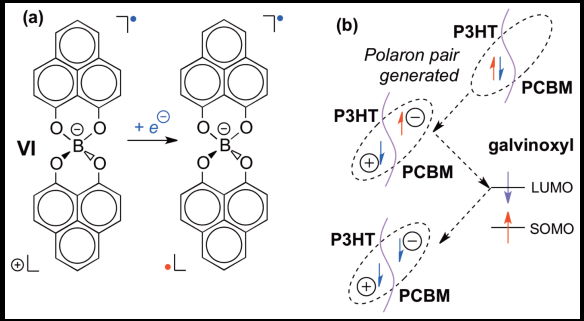
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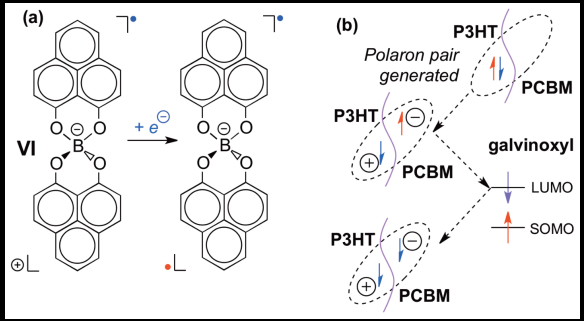
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Non-aufbau by electron addition (Figure 17)



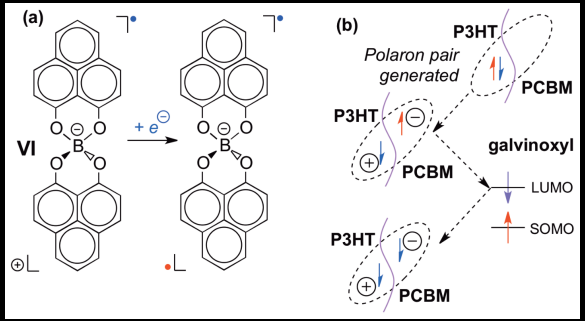
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- self-doping: in $TTFCOO^{-}NH_4^{+}$ the SOMO of $TTF^{+\bullet}$ is not the highest occupied orbital

Part XVI

Broken symmetry

XVI

Cheap methods for diradicals

The paper says

> "relatively 'cheap' methods such as unrestricted or broken symmetry formalisms suffer from a number of serious drawbacks, while rigorous methods … are generally not feasible for systems large enough to be practically valuable." (p. 443)

broken-symmetry (BS) determinant $|a'\bar{b}'\rangle$, a' on centre l, b' on r, overlap $s = \langle a'|b'\rangle$

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a single determinant cannot be the open-shell singlet (Part III)

Yamaguchi's projection, derived

Assumptions

two-centre Heisenberg model $\hat{H} = -2J\,\hat{\mathbf{S}}_1\cdot\hat{\mathbf{S}}_2$; BS state $= c_S\Psi_S + c_T\Psi_T$ ($M_S = 0$)

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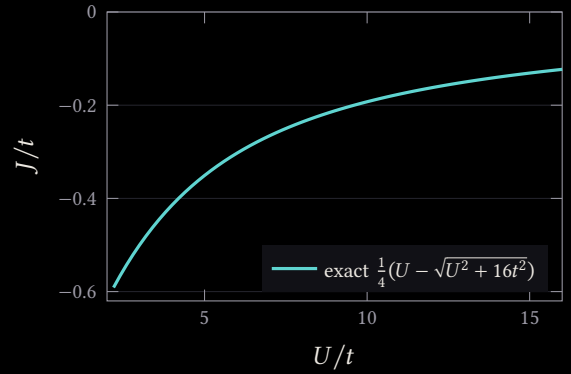
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Lean: `yamaguchi`; sign conventions differ between sources, here $J < 0$ is antiferromagnetic

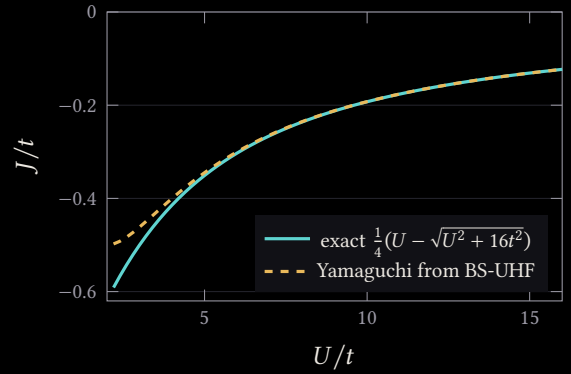
How good is it? The Hubbard dimer

T14: self-consistent BS-UHF vs the exact singlet--triplet gap



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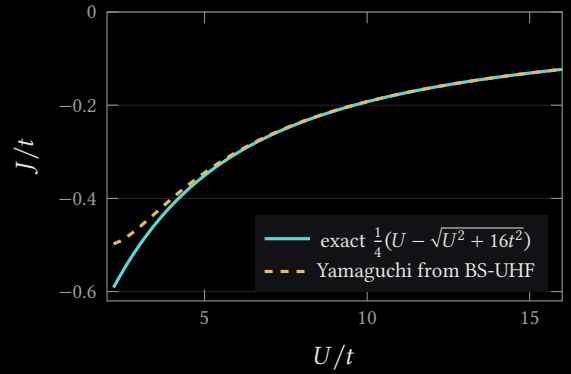
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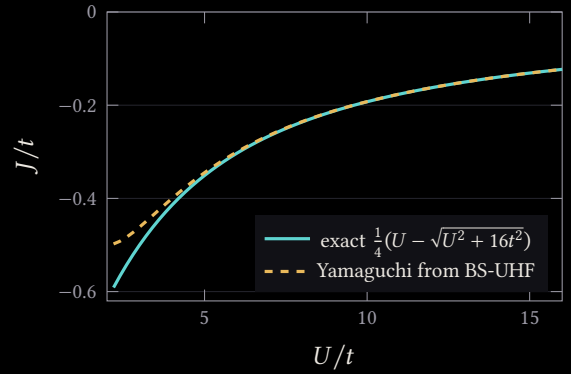
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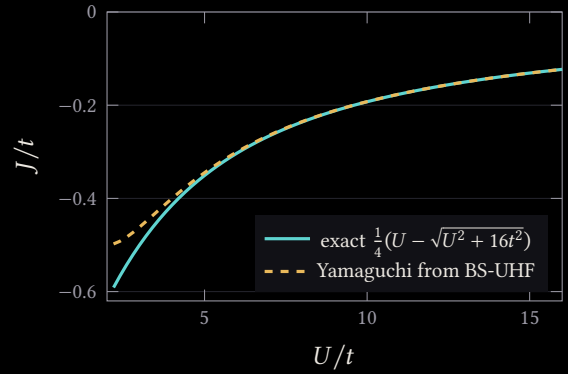
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VERIFIED "serious drawbacks" is fair

Part XVII

Our verification

XVII

The repository

`CH5103_Term_Paper/`	
`paper/`	the review (PDF)
`derivations/`	18 LaTeX sections
`verification/`	T1–T15 scripts, logs
`lean_proofs/`	Lean 4 + Mathlib (T17)
`assets/`	figures, captions (JSONL)
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`Makefile`	one command per artefact

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make derivations
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Python 3.12 pinned by `uv.lock`
PySCF 2.14, SymPy 1.14
Lean 4.29.1, Mathlib 4.29.1

The test matrix (1/2)

test	what	method	result
T1	spin functions, Fig. 1 combinations, sign erratum	SymPy	VERIFIED
T2	$2e/2o$ Hamiltonian vs full CI (5 random sets)	PySCF FCI	VERIFIED
T3	$K_{pq} \geq 0$, $J_{pq} \geq K_{pq}$	RHF/6-31G H_2O	VERIFIED
T4	He $2\,^3S/2\,^1S$: virial, less screening	exact 2-electron	VERIFIED
T5	H_2 dissociation limits	STO-3G closed forms	VERIFIED
T6	two-orbital gap model	SymPy	VERIFIED
T7	kinetic exchange, extended dimer	SymPy + FCI	VERIFIED
T8	Hückel NBMO rules, disjointness	NumPy	VERIFIED
T9	Ovchinnikov/Lieb spins; spin alternation	exact diagonalisation	VERIFIED

The test matrix (2/2)

test	what	method	result
T9b	gap $\propto U$ vs $\propto U^2$	exact diagonalisation	VERIFIED
T10	S–T gap = HOMO difference	SymPy	VERIFIED
T11a	ROHF stationarity, canonicalisation	ROHF/6-31G*	VERIFIED
T11b	distonic anion, B3LYP	UB3LYP ΔSCF	ERROR
T11c	distonic anions/acids, CAM-B3LYP	UKS ΔSCF	VERIFIED
T12	3d/4s identity	SymPy	VERIFIED
T13	$[\hat{l}_1, r_{12}^{-1}] \neq 0$, $[\hat{L}, r_{12}^{-1}] = 0$	SymPy	VERIFIED
T14	Yamaguchi projection	BS-UHF dimer	VERIFIED
T15	same- vs opposite-spin correlation	MP2, He	VERIFIED
T17	17 algebraic theorems	Lean 4 + Mathlib	VERIFIED

T11b "fails" honestly: B3LYP misses the crossover by 0.07 eV; T11c explains why

Inside T4: helium solved exactly

verification/t03_t05_hund_virial_h2.py (excerpt, unedited)

```
    # Exact two-electron problem in the spatial pair space Psi(1,2) = sum_pq C_pq phi_p(1) phi_q(2):
    #   H[(p,q),(r,s)] = h_pr d_qs + d_pr h_qs + (pr|qs)
    # singlet <-> C symmetric, triplet <-> C antisymmetric (spin factor carries the other symmetry).
    I = np.eye(n)
    H = (np.einsum("pr,qs->pqrs", h, I) + np.einsum("pr,qs->pqrs", I, h)
         + eri.transpose(0, 2, 1, 3)).reshape(n * n, n * n)
    sign = +1 if which == "1S" else -1
    cols = []
    for p in range(n):
        for q in range(p if sign > 0 else p + 1, n):
            v = np.zeros((n, n)); v[p, q] += 1; v[q, p] += sign
            cols.append(v.ravel() / np.linalg.norm(v))
    B = np.array(cols).T
    w, Y = np.linalg.eigh(B.T @ H @ B)
    pick = 1 if which == "1S" else 0          # 1S: root 0 is 1^1S, root 1 is 2^1S ; 3S: root 0 is 2^3S
    e = w[pick]; C = (B @ Y[:, pick]).reshape(n, n)
    Tex = 2 * np.einsum("pq,pr,rq", C, T, C)
    Vne = 2 * np.einsum("pq,pr,rq", C, V, C)
    Vee = np.einsum("pq,prqs,rs", C, eri, C)
    assert np.isclose(Tex + Vne + Vee, e, atol=1e-8)
    return e, Tex, Vne, Vee
```

symmetric C ↔ singlet, antisymmetric C ↔ triplet (Part III)

Inside T11a: ROHF Fock blocks

verification/t11_rohf_and_distonic.py (excerpt)

```python
def rohf_blocks(atom, spin, charge=0, basis="6-31g*"):
    mol = gto.M(atom=atom, spin=spin, charge=charge, basis=basis, verbose=0)
    mf = scf.ROHF(mol).run(conv_tol=1e-11)
    dm = mf.make_rdm1()
    vhf = mf.get_veff(mol, dm)
    h1 = mf.get_hcore()
    C = mf.mo_coeff
    Fa = C.T @ (h1 + vhf[0]) @ C; Fb = C.T @ (h1 + vhf[1]) @ C
    occ = mf.mo_occ
    c, o, v = np.where(occ == 2)[0], np.where(occ == 1)[0], np.where(occ == 0)[0]
    return mol, mf, Fa, Fb, c, o, v

    g_co = np.abs(Fb[np.ix_(c, o)]).max(); g_ov = np.abs(Fa[np.ix_(o, v)]).max()
    g_cv = np.abs((Fa + Fb)[np.ix_(c, v)]).max()
    n_co = np.abs(Fa[np.ix_(c, o)]).max(); n_ov = np.abs(Fb[np.ix_(o, v)]).max()
    stat = max(g_co, g_ov, g_cv) < 1e-5 and max(n_co, n_ov) > 1e-3
```

UHF Fock matrices from the ROHF densities, then the blocks the stationarity table says must vanish

Inside T17: Lean proves the less-screening logic

verification/lean_proofs/Ch5103/Identities.lean (excerpt)

```
/-- Less screening: if the triplet is lower (ΔE < 0), both states obey the virial theorem and the
triplet has the larger e-e repulsion (ΔVee > 0), then ΔT > 0 and ΔVne < 2ΔE < 0 with |ΔVne| > ΔVee. -/
theorem less_screening (dT dVne dVee dE : ℝ)
    (hvir : 2 * dT + (dVne + dVee) = 0) (hE : dE = dT + dVne + dVee)
    (hlow : dE < 0) (hee : 0 < dVee) :
    0 < dT ∧ dVne < 2 * dE ∧ dVne < 0 ∧ dVee < -dVne := by
  refine ⟨by linarith, by linarith, by linarith, by linarith⟩

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Axiom audit

`#print axioms`, all 17 theorems: only `propext`, `Classical.choice`, `Quot.sound`

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writing the test first is what turned "I believe" into "checked"

Part XVIII

Verdict

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ROHF orbital energies are convention-dependent	VERIFIED OVERSTATED

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Katriel & Pauncz cited as 1975 (it is 1977)	ERROR

The errors, once more

Sign error, p. 441

$\frac{1}{2}(D_{\mathrm{iii}} + D_{\mathrm{iv}})$ is the $M_S = 0$ triplet; the open-shell singlet is $\frac{1}{2}(D_{\mathrm{iv}} - D_{\mathrm{iii}})$

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none of these touches a conclusion: the chemistry of the review stands

Take-home

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Even the open-shell singlet in print can fluctuate.

Key references

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full access status of every source: derivations, §17

Thank you

derivations • code • proofs • slides

`github.com/Shuvam-Banerji-Seal/CH5103_Term_Paper`